05 : WMAP9 + DESI DR2 + Pantheon+ Joint $w_0 w_a$ CDM Fitting (Cobaya)

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Extension of Notebook 04 from flat ΛCDM to dynamical dark energy (CPL), $w(a) = w_0 + w_a(1-a)$, sampled with **Cobaya** instead of emcee.

Joint likelihood

$$\ln L_{total}(\theta) = \ln L_{WMAP9}(\theta) + \ln L_{DR2}(\theta) + \ln L_{Pantheon+}(\theta)$$

θ = [$\Omega_b h^2$, $\Omega_c h^2$, H_0, n_s, $\ln(10^{10} A_s)$, τ, w_0, w_a]

WMAP9: your official ctypes likelihood (Notebook 04, Cell 2), wrapped as a Cobaya likelihood.

DESI DR2 BAO: Cobaya's official `bao.desi_dr2.desi_bao_all` (DESI collaboration likelihood files).

Pantheon+: Cobaya's official `sn.pantheonplus`.

Key rule

No Planck r_d prior and no Planck data. WMAP9 calibrates r_d through the acoustic peaks.

Prior bounds : widened from Notebook 04

$H_0 \in [50, 90]$, $w_0 \in [-3, 1]$, $w_a \in [-3, 2]$ (DESI convention), $w_0 + w_a < 0$.

Cell 1 : Imports and Paths

In [ ]:
```
import numpy as np
import matplotlib.pyplot as plt
import os, ctypes, warnings
warnings.filterwarnings('ignore')
import camb
import cobaya
from cobaya.model import get_model
from cobaya.run import run
from cobaya.likelihood import Likelihood
```

```
from scipy import stats
from pathlib import Path

ROOT         = os.path.expanduser('~/Desktop/Cosmology')
DATA_DIR     = ROOT + '/data'
LIKE_DIR     = DATA_DIR + '/wmap9/wmap_likelihood_v5'
SO_PATH      = DATA_DIR + '/wmap9/libwmap9.so'
PACKAGES     = DATA_DIR + '/cobaya_packages'                    # D
ANALYSIS_DIR = ROOT + '/analyses/wmap9_desi_dr2_pantheon_w0wa'
CHAIN_DIR    = ANALYSIS_DIR + '/chains'
IMG_DIR      = ANALYSIS_DIR + '/figures'
RESULTS_DIR  = ANALYSIS_DIR + '/results'
REF_DIR      = ANALYSIS_DIR + '/reference_chains'
for d in (CHAIN_DIR, IMG_DIR, RESULTS_DIR):
    os.makedirs(d, exist_ok=True)
import sys
sys.path.insert(0, ANALYSIS_DIR)

# Cobaya parameter names: logA = ln(10^10 A_s), w = w0
PARAM_NAMES  = ['ombh2', 'omch2', 'H0', 'ns', 'logA', 'tau', 'w', '
PARAM_LABELS = [r'$\Omega_b h^2$', r'$\Omega_c h^2$', r'$H_0$',
                r'$n_s$', r'$\ln(10^{10}A_s)$', r'$\tau$', r'$w_0$'

PRIOR_BOUNDS = {
    'ombh2': (0.018, 0.028),
    'omch2': (0.08,  0.16 ),
    'H0':    (50.,   90.  ),   # widened from [55, 85]
    'ns':    (0.9,   1.05 ),
    'logA':  (2.9,   3.3  ),
    'tau':   (0.04,  0.20 ),
    'w':     (-3.0,  1.0  ),   # w0
    'wa':    (-3.0,  2.0  ),
}

# Data combination for this run
USE_SN = True        # True = WMAP9 + DESI DR2 + Pantheon+ ; False

DATA_LABEL = 'WMAP9 + DESI DR2' + (' + Pantheon+' if USE_SN else ''
TAG        = 'wmap9_dr2' + ('_pantheonplus' if USE_SN else '')

print('Paths:')
for k, v in [('LIKE_DIR', LIKE_DIR), ('SO_PATH', SO_PATH), ('PACKAG
    status = 'OK' if os.path.exists(v) else 'NOT FOUND'
    print(f'  {k:12s}: [{status}]  {v}')
print('CAMB version  :', camb.__version__)
print('Cobaya version:', cobaya.__version__)
print('Data          :', DATA_LABEL)
```

```
Paths:
  LIKE_DIR    : [OK]  /Users/cherianpi/Desktop/Cosmology/data/wmap9/
wmap_likelihood_v5
  SO_PATH     : [OK]  /Users/cherianpi/Desktop/Cosmology/data/wmap9/
libwmap9.so
  PACKAGES    : [OK]  /Users/cherianpi/Desktop/Cosmology/data/cobaya
_packages
  ANALYSIS_DIR: [OK]  /Users/cherianpi/Desktop/Cosmology/analyses/wm
ap9_desi_dr2_pantheon_w0wa
CAMB version  : 1.5.5
Cobaya version: 3.5.7
Data          : WMAP9 + DESI DR2 + Pantheon+
```

Cell 2 : Load WMAP9 Likelihood Library

Same ctypes wrapper as Notebook 04, Cell 2, now kept in `wmap9_likelihood.py` next to this notebook. Cobaya saves a copy of every likelihood with its chains, which works for a class imported from a file but not for one defined inside a notebook.

In [4]:
```
import wmap9_likelihood as W
from wmap9_likelihood import wmap_like, WMAP9, COMP_NAMES

W.load_library()
print('WMAP9 likelihood library loaded and initialised:', W.SO_PATH

# Quick verification
tc = np.loadtxt(LIKE_DIR + '/data/test_cls_v5.dat')
lt, comps = wmap_like(tc[:,1], tc[:,4], tc[:,2], tc[:,3])
print(f'Verification -2lnL = {lt:.4f}  (expect 7557.97)')
for n, c in zip(COMP_NAMES, comps):
    print(f'  {n:24s} {c:12.4f}')
```

```
 Initializing WMAP likelihood, version v5
WMAP9 likelihood library loaded and initialised: /Users/cherianpi/De
sktop/Cosmology/data/wmap9/libwmap9.so
Verification -2lnL = 7557.9660  (expect 7557.97)
  TT high-l MASTER            1200.0052
  TT low-l Gibbs               -13.6149
  TT low-l det                   0.0000
  Beam+ptsrc                     0.7356
  TE high-l chi2               815.4338
  TE high-l det               3541.5372
  lowl TT/TE/EE/BB chi2       1320.9946
  lowl TT/TE/EE/BB det         692.8745
```

Cell 3 : WMAP9 as a Cobaya Likelihood

`WMAP9` (in `wmap9_likelihood.py`) asks Cobaya's CAMB for the lensed spectra and hands D_ℓ (μK^2, $\ell = 2..1200$) to the same `wmap_like` as Notebook 04. The CAMB settings match Notebook 04: one massive neutrino

of 0.06 eV, lensed spectra, `lens_potential_accuracy = 0`, plus PPF dark energy.

In [5]:
```
CAMB_THEORY = {'camb': {'extra_args': {'num_massive_neutrinos': 1,
                                       'dark_energy_model': 'ppf',
                                       'lens_potential_accuracy': 0
print('WMAP9 Cobaya likelihood:', WMAP9.__module__ + '.' + WMAP9.__
print('Requirements          :', WMAP9.get_requirements(None))
```

```
WMAP9 Cobaya likelihood: wmap9_likelihood.WMAP9
Requirements          : {'Cl': {'tt': 1200, 'te': 1200, 'ee': 1200,
'bb': 1200}}
```

Cell 4 : Data: DESI DR2 BAO and Pantheon+

Both come from the official Cobaya likelihoods installed in `data/cobaya_packages`. This cell only reads and prints the files so the inputs are visible; Cobaya does the actual loading.

In [6]:
```
bao_dir = Path(PACKAGES) / 'data' / 'bao_data' / 'desi_bao_dr2'
print('DESI DR2 BAO (desi_gaussian_bao_ALL_GCcomb_mean.txt):')
with open(bao_dir / 'desi_gaussian_bao_ALL_GCcomb_mean.txt') as f:
    rows = [l.split() for l in f if l.strip() and not l.startswith(
cov = np.loadtxt(bao_dir / 'desi_gaussian_bao_ALL_GCcomb_cov.txt')
for (z, v, q), e in zip(rows, np.sqrt(np.diag(cov))):
    print(f'  z = {float(z):5.3f}   {q:11s} {float(v):9.4f} ± {e:.4
print(f'  {len(rows)} measurements')

sn_dir = Path(PACKAGES) / 'data' / 'sn_data' / 'PantheonPlus'
with open(sn_dir / 'Pantheon+SH0ES.dat') as f:
    header = f.readline().split()
    zhd = np.array([float(l.split()[header.index('zHD')]) for l in
print(f'\nPantheon+ : {len(zhd)} SNe in the file, {np.sum(zhd > 0.0
      f'z up to {zhd.max():.3f}')
```

```
DESI DR2 BAO (desi_gaussian_bao_ALL_GCcomb_mean.txt):
  z = 0.295   DV_over_rs     7.9417 ± 0.0761
  z = 0.510   DM_over_rs    13.5876 ± 0.1684
  z = 0.510   DH_over_rs    21.8629 ± 0.4289
  z = 0.706   DM_over_rs    17.3507 ± 0.1799
  z = 0.706   DH_over_rs    19.4553 ± 0.3339
  z = 0.934   DM_over_rs    21.5756 ± 0.1618
  z = 0.934   DH_over_rs    17.6415 ± 0.2010
  z = 1.321   DM_over_rs    27.6009 ± 0.3246
  z = 1.321   DH_over_rs    14.1760 ± 0.2246
  z = 1.484   DM_over_rs    30.5119 ± 0.7636
  z = 1.484   DH_over_rs    12.8170 ± 0.5180
  z = 2.330   DH_over_rs     8.6315 ± 0.1011
  z = 2.330   DM_over_rs    38.9890 ± 0.5317
  13 measurements

Pantheon+ : 1701 SNe in the file, 1590 used (zHD > 0.01), z up to 2.
261
```

Cell 5 : Cobaya Model

Parameters, priors and the dark-energy prior $w_0 + w_a < 0$. `ref` is where chains and the minimiser start (the Notebook 04 joint MAP region), `proposal` the initial step size (Cobaya then learns the proposal covariance). Ω_m and r_d are derived.

In [7]:
```
REF = {'ombh2': (0.02253, 0.0002), 'omch2': (0.11515, 0.002), 'H0':
       'ns': (0.972, 0.005), 'logA': (3.096, 0.01), 'tau': (0.088,
       'w': (-0.9, 0.05), 'wa': (-0.3, 0.1)}
LATEX = {'ombh2': r'\Omega_b h^2', 'omch2': r'\Omega_c h^2', 'H0':
         'logA': r'\ln(10^{10}A_s)', 'tau': r'\tau', 'w': r'w_0', '

def make_params(model='w0wa'):
    params = {}
    for name in PARAM_NAMES:
        lo, hi = PRIOR_BOUNDS[name]
        loc, sc = REF[name]
        params[name] = {'prior': {'min': lo, 'max': hi},
                        'ref': {'dist': 'norm', 'loc': loc, 'scale'
                        'proposal': sc, 'latex': LATEX[name]}
    params['logA']['drop'] = True
    params['As'] = {'value': 'lambda logA: 1e-10 * np.exp(logA)', '
    params['mnu'] = 0.06
    params['omk'] = 0.0
    params['omegam'] = {'latex': r'\Omega_m'}     # derived
    params['rdrag']  = {'latex': r'r_d'}          # derived
    if model == 'lcdm':
        params['w'], params['wa'] = -1.0, 0.0
    return params

def make_info(model='w0wa'):
    likelihood = {'wmap9': WMAP9, 'bao.desi_dr2.desi_bao_all': None
    if USE_SN:
```

```
        likelihood['sn.pantheonplus'] = None
    info = {'packages_path': PACKAGES, 'theory': CAMB_THEORY,
            'likelihood': likelihood, 'params': make_params(model)}
    if model == 'w0wa':
        info['prior'] = {'early_dark_energy_bound': 'lambda w, wa:
    return info

model = get_model(make_info('w0wa'))
print('Sampled parameters:', list(model.parameterization.sampled_pa
print('Derived parameters:', list(model.parameterization.derived_pa
print('Likelihoods       :', list(model.likelihood))
```

```
[prior] *WARNING* External prior 'early_dark_energy_bound' loaded. M
ind that it might not be normalized!
[camb] `camb` module loaded successfully from /opt/anaconda3/envs/Ca
mb/lib/python3.11/site-packages/camb
[camb] `camb` module loaded successfully from /opt/anaconda3/envs/Ca
mb/lib/python3.11/site-packages/camb
[bao.desi_dr2.desi_bao_all] Initialized.
Sampled parameters: ['ombh2', 'omch2', 'H0', 'ns', 'logA', 'tau', '
w', 'wa']
Derived parameters: ['As', 'omegam', 'rdrag', 'chi2__BAO', 'chi2__S
N']
Likelihoods       : ['wmap9', 'bao.desi_dr2.desi_bao_all', 'sn.panth
eonplus']
```

Cell 5b : Validation: ΛCDM limit against Notebook 04

At $w_0 = -1$, $w_a = 0$, Cobaya's CAMB + `WMAP9` must reproduce the Notebook 04 WMAP9 $-2 \ln L$ at the test point (Notebook 04 / Notebook 00: 7560.42). Small differences ($\lesssim 0.1$) can come from Cobaya's own CAMB accuracy and ℓ_{max} choices. A large one means a settings mismatch.

In [8]:
```
def get_cls_nb04(ombh2, omch2, H0, ns, ln10As, tau):
    """Notebook 04 CAMB function, unchanged."""
    cp = camb.CAMBparams()
    cp.set_cosmology(H0=H0, ombh2=ombh2, omch2=omch2, tau=tau)
    cp.InitPower.set_params(ns=ns, As=np.exp(ln10As) * 1e-10)
    cp.set_for_lmax(1250, lens_potential_accuracy=0)
    Dl = camb.get_results(cp).get_cmb_power_spectra(cp, CMB_unit='m
    ell = np.arange(2, 1201)
    return Dl[ell, 0], Dl[ell, 3], Dl[ell, 1], Dl[ell, 2]

p_test = dict(ombh2=0.02264, omch2=0.1138, H0=70.0, ns=0.972, logA=
loglikes, derived = model.loglikes(p_test, as_dict=True)
neg2_cobaya = -2 * loglikes['wmap9']
neg2_nb04   = wmap_like(*get_cls_nb04(0.02264, 0.1138, 70.0, 0.972,
print(f'WMAP9 -2lnL  Cobaya (PPF, w0=-1, wa=0) = {neg2_cobaya:.4f}'
print(f'WMAP9 -2lnL  Notebook 04 function      = {neg2_nb04:.4f}')
print(f'difference                             = {neg2_cobaya - neg
```

```
WMAP9 -2lnL  Cobaya (PPF, w0=-1, wa=0) = 7560.4320
WMAP9 -2lnL  Notebook 04 function      = 7560.4215
difference                             = +0.0104
```

Cell 6 : Individual Log-Likelihoods

In [9]:
```
theta_test = dict(ombh2=0.02253, omch2=0.11515, H0=68.89, ns=0.972,
loglikes, derived = model.loglikes(theta_test, as_dict=True)
for name, ll in loglikes.items():
    print(f'{name:28s} lnL = {ll:12.3f}   chi2 = {-2*ll:10.3f}')
print(f'Derived: Omega_m = {derived["omegam"]:.4f},  r_d = {derived
```

```
wmap9                        lnL =    -3778.996   chi2 =   7557.992
bao.desi_dr2.desi_bao_all    lnL =       -5.436   chi2 =     10.871
sn.pantheonplus              lnL =     -704.015   chi2 =   1408.031
Derived: Omega_m = 0.2915,  r_d = 148.220 Mpc
```

Cell 7 : Joint Prior and Posterior

In [ ]:
```
lp = model.logposterior(theta_test)
print(f'Joint log-posterior at the test point: {lp.logpost:.3f}')
print(f'  = log-prior {lp.logprior:.3f} + log-likelihood {lp.loglik

# w0 + wa < 0
bad = dict(theta_test, w=-0.5, wa=1.0)
print(f'Point with w0 + wa = {bad["w"] + bad["wa"]:+.1f}: log-prior
```

```
Joint log-posterior at the test point: -4483.354
  = log-prior 5.092 + log-likelihood -4488.447
Point with w0 + wa = +0.5: log-prior = -inf
```

Cell 8 : MAP Fits (w_0w_a and ΛCDM on the same data)

Cobaya's `minimize`, run twice from different starting points:

- **ΛCDM**: $w_0 = -1$, $w_a = 0$ fixed;
- w_0w_a: all 8 parameters free.

$\Delta\chi^2 = \chi^2_{\Lambda CDM} - \chi^2_{w_0w_a}$ is the headline number for any preference for dynamical dark energy. Results are written to `results/`.

In [ ]:
```
def read_minimum(prefix):
    """Cobaya's minimum as a dict {column: value} (from <prefix>.mi
    with open(prefix + '.minimum.txt') as f:
        names = f.readline().lstrip('#').split()
        vals  = f.readline().split()
    out = {n: float(v) for n, v in zip(names, vals)}
    out.setdefault('w', -1.0)
    out.setdefault('wa', 0.0)
    return out

def run_minimize(model_name):
```

```
    prefix = RESULTS_DIR + f'/{TAG}_{model_name}_map'
    if os.path.exists(prefix + '.minimum.txt'):          # already
        print(f'Loading existing minimum: {prefix}.minimum.txt')
        return read_minimum(prefix)
    info = make_info(model_name)
    info['sampler'] = {'minimize': {'method': 'bobyqa', 'best_of':
    info['output']  = prefix
    run(info)
    return read_minimum(prefix)

best = {m: run_minimize(m) for m in ('lcdm', 'w0wa')}

like_keys = ['chi2__' + k for k in make_info('w0wa')['likelihood']]
print(f'\n=== MAP — {DATA_LABEL} ===')
print(f'{"Param":<10} {"LCDM":>11} {"w0wa":>11}')
print('-' * 34)
for name in PARAM_NAMES + ['omegam', 'rdrag']:
    print(f'{name:<10} {best["lcdm"][name]:11.5f} {best["w0wa"][nam
print(f'\n{"chi2":<30} {"LCDM":>11} {"w0wa":>11}')
for k in like_keys + ['chi2']:
    print(f'{k:<30} {float(best["lcdm"][k]):11.3f} {float(best["w0w

dchi2 = float(best['lcdm']['chi2']) - float(best['w0wa']['chi2'])
p_val = stats.chi2.sf(max(dchi2, 0), 2)
sigma = stats.norm.isf(p_val / 2) if dchi2 > 0 else 0.0
print(f'\nDelta chi2 (LCDM - w0wa) = {dchi2:.2f}  ->  {sigma:.2f} s
```

```
[output] Output to be read-from/written-into folder '/Users/cherianp
i/Desktop/Cosmology/analyses/wmap9_desi_dr2_pantheon_w0wa/results',
with prefix 'wmap9_dr2_pantheonplus_lcdm_map'
[camb] `camb` module loaded successfully from /opt/anaconda3/envs/Ca
mb/lib/python3.11/site-packages/camb
[bao.desi_dr2.desi_bao_all] Initialized.
[minimize] Initializing
[minimize] Run 1/2 will start from random initial point:
[minimize] {'ombh2': np.float64(0.022822674141862433), 'omch2': np.f
loat64(0.11551545796608456), 'H0': np.float64(69.17869568870502), 'n
s': np.float64(0.9684347090061712), 'logA': np.float64(3.10124098644
50876), 'tau': np.float64(0.08888857400654682)}
[minimize] Run 2/2 will start from random initial point:
[minimize] {'ombh2': np.float64(0.02258644174084354), 'omch2': np.fl
oat64(0.1144915998734857), 'H0': np.float64(68.7526779534089), 'ns':
np.float64(0.9669617508897657), 'logA': np.float64(3.10860557750017
1), 'tau': np.float64(0.0855143659804776)}
[minimize] Starting run 1/2
[minimize] Run 1/2 converged.
[minimize] Starting run 2/2
[minimize] Run 2/2 converged.
[minimize] Finished successfully!
[minimize] -log(posterior) minimized to 4479.64
[minimize] Parameter values at minimum:
   weight  minuslogpost     ombh2     omch2         H0        ns
logA       tau            As    omegam       rdrag  chi2__BAO     ch
i2__SN  minuslogprior  minuslogprior__0         chi2  chi2__wmap9  c
hi2__bao.desi_dr2.desi_bao_all  chi2__sn.pantheonplus
0     1.0   4479.637402  0.022505  0.116292  68.578762  0.970231  3.
```

```
095827  0.085478  2.210551e-09  0.296493  147.939748  10.563592  140
6.797713      -8.088012         -8.088012  8975.450827  7558.089522
10.563592            1406.797713
[minimize] Full set of minima:
{'0': (4479.706145900909, True), '1': (4479.63740198626, True)}
[output] Output to be read-from/written-into folder '/Users/cherianp
i/Desktop/Cosmology/analyses/wmap9_desi_dr2_pantheon_w0wa/results',
with prefix 'wmap9_dr2_pantheonplus_w0wa_map'
[prior] *WARNING* External prior 'early_dark_energy_bound' loaded. M
ind that it might not be normalized!
[camb] `camb` module loaded successfully from /opt/anaconda3/envs/Ca
mb/lib/python3.11/site-packages/camb
[bao.desi_dr2.desi_bao_all] Initialized.
[minimize] Initializing
[minimize] Run 1/2 will start from random initial point:
[minimize] {'ombh2': np.float64(0.02244975286769502), 'omch2': np.fl
oat64(0.11636484310411893), 'H0': np.float64(67.84411656843305), 'n
s': np.float64(0.9794614553930628), 'logA': np.float64(3.11142219727
75464), 'tau': np.float64(0.08423303307955368), 'w': np.float64(-0.9
131691872535658), 'wa': np.float64(-0.4077083443823814)}
[minimize] Run 2/2 will start from random initial point:
[minimize] {'ombh2': np.float64(0.022366839802019776), 'omch2': np.f
loat64(0.11488086824409917), 'H0': np.float64(68.99108496262974), 'n
s': np.float64(0.9697359878957652), 'logA': np.float64(3.11292186841
5183), 'tau': np.float64(0.08830314757354185), 'w': np.float64(-0.87
02811246566102), 'wa': np.float64(-0.24979689320460713)}
[minimize] Starting run 1/2
[minimize] Run 1/2 converged.
[minimize] Starting run 2/2
[minimize] Run 2/2 converged.
[minimize] Finished successfully!
[minimize] -log(posterior) minimized to 4479.67
[minimize] Parameter values at minimum:
   weight  minuslogpost     ombh2     omch2         H0        ns
logA       tau         w        wa            As    omegam      rdra
g  chi2__BAO     chi2__SN  minuslogprior  minuslogprior__0  minuslog
prior__early_dark_energy_bound         chi2  chi2__wmap9  chi2__bao.
desi_dr2.desi_bao_all  chi2__sn.pantheonplus
0     1.0   4479.669632  0.022689  0.116426  67.499783  0.973717  3.
103578  0.088171 -0.887466 -0.260995  2.227753e-09  0.306746  147.69
717   8.689316  1402.608704      -5.092279         -5.092279
-0.0  8969.523822  7558.225801                          8.689316
1402.608704
[minimize] Full set of minima:
{'0': (4479.669631588704, True), '1': (4479.7465654582165, True)}

=== MAP — WMAP9 + DESI DR2 + Pantheon+ ===
Param             LCDM        w0wa
----------------------------------
ombh2          0.02250     0.02269
omch2          0.11629     0.11643
H0            68.57876    67.49978
ns             0.97023     0.97372
logA           3.09583     3.10358
tau            0.08548     0.08817
w             -1.00000    -0.88747
```

```
wa              0.00000    -0.26100
omegam          0.29649     0.30675
rdrag         147.93975   147.69717

chi2                                    LCDM        w0wa
chi2__wmap9                         7558.090    7558.226
chi2__bao.desi_dr2.desi_bao_all       10.564       8.689
chi2__sn.pantheonplus               1406.798    1402.609
chi2                                8975.451    8969.524

Delta chi2 (LCDM - w0wa) = 5.93  ->  1.95 sigma for 2 extra paramete
rs
```

Cell 9 : MCMC Sampling (Cobaya)

It stops when the Gelman–Rubin $R-1<0.01$ for the means and $R-1<0.2$ for the 95% bounds. Progress is written to `chains/` continuously.

In [ ]:
```python
info_mcmc = make_info('w0wa')
info_mcmc['sampler'] = {'mcmc': {
    'Rminus1_stop': 0.01,
    'Rminus1_cl_stop': 0.2,
    'learn_proposal': True,
    'max_tries': 2000,
    'measure_speeds': False,
}}
info_mcmc['output'] = CHAIN_DIR + f'/{TAG}_w0wa'
info_mcmc['resume'] = True

print(f'Cobaya MCMC : {DATA_LABEL}, w0waCDM')
print(f'Output      : {info_mcmc["output"]}.*')
updated_info, mcmc = run(info_mcmc)
print('Done.')
```

```
Cobaya MCMC : WMAP9 + DESI DR2 + Pantheon+, w0waCDM
Output      : /Users/cherianpi/Desktop/Cosmology/analyses/wmap9_desi
_dr2_pantheon_w0wa/chains/wmap9_dr2_pantheonplus_w0wa.*
[output] Output to be read-from/written-into folder '/Users/cherianp
i/Desktop/Cosmology/analyses/wmap9_desi_dr2_pantheon_w0wa/chains', w
ith prefix 'wmap9_dr2_pantheonplus_w0wa'
[prior] *WARNING* External prior 'early_dark_energy_bound' loaded. M
ind that it might not be normalized!
[camb] `camb` module loaded successfully from /opt/anaconda3/envs/Ca
mb/lib/python3.11/site-packages/camb
[bao.desi_dr2.desi_bao_all] Initialized.
[mcmc] Getting initial point... (this may take a few seconds)
[mcmc] Initial point: ombh2:0.02231854, omch2:0.1187989, H0:68.8874
6, ns:0.9692658, logA:3.101842, tau:0.08637946, w:-0.9120842, wa:-0.
3182238
[mcmc] Covariance matrix not present. We will start learning the cov
ariance of the proposal earlier: R-1 = 30 (would be 2 if all params
loaded).
```

```
[mcmc] Sampling!
[mcmc] Progress @ 2026-11-07 12:58:32 : 1 steps taken, and 0 accepte
d.
[mcmc] Progress @ 2026-11-07 12:59:32 : 90 steps taken, and 13 accep
ted.
[mcmc] Progress @ 2026-11-07 13:00:32 : 190 steps taken, and 28 acce
pted.
[mcmc] Progress @ 2026-11-07 13:01:33 : 284 steps taken, and 39 acce
pted.
[mcmc] Progress @ 2026-11-07 13:02:33 : 382 steps taken, and 51 acce
pted.
[mcmc] Progress @ 2026-11-07 13:03:33 : 474 steps taken, and 64 acce
pted.
[mcmc] Progress @ 2026-11-07 13:04:34 : 570 steps taken, and 80 acce
pted.
[mcmc] Progress @ 2026-11-07 13:05:34 : 667 steps taken, and 91 acce
pted.
[mcmc] Progress @ 2026-11-07 13:06:35 : 761 steps taken, and 104 acc
epted.
[mcmc] Progress @ 2026-11-07 13:07:35 : 859 steps taken, and 118 acc
epted.
[mcmc] Progress @ 2026-11-07 13:08:35 : 948 steps taken, and 135 acc
epted.
[mcmc] Progress @ 2026-11-07 13:09:36 : 1044 steps taken, and 151 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:10:36 : 1139 steps taken, and 161 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:11:37 : 1231 steps taken, and 171 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:12:37 : 1332 steps taken, and 186 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:13:37 : 1426 steps taken, and 202 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:14:38 : 1521 steps taken, and 218 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:15:38 : 1610 steps taken, and 228 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:16:38 : 1706 steps taken, and 241 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:17:39 : 1798 steps taken, and 257 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:18:39 : 1886 steps taken, and 265 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:19:40 : 1977 steps taken, and 273 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:20:40 : 2071 steps taken, and 284 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:21:40 : 2178 steps taken, and 303 ac
cepted.
[mcmc] Learn + convergence test @ 320 samples accepted.
[mcmc]  - Acceptance rate: 0.142
[mcmc]  - Convergence of means: R-1 = 3.424295 after 256 accepted st
eps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 13:22:41 : 2277 steps taken, and 321 ac
cepted.
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[mcmc] Progress @ 2026-11-07 13:23:41 : 2381 steps taken, and 361 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:24:41 : 2489 steps taken, and 402 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:25:41 : 2598 steps taken, and 451 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:26:41 : 2699 steps taken, and 498 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:27:41 : 2801 steps taken, and 536 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:28:41 : 2908 steps taken, and 573 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:29:41 : 3015 steps taken, and 626 ac
cepted.
[mcmc] Learn + convergence test @ 640 samples accepted.
[mcmc]  - Acceptance rate: 0.239
[mcmc]  - Convergence of means: R-1 = 0.978547 after 512 accepted st
eps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 13:30:42 : 3118 steps taken, and 661 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:31:42 : 3227 steps taken, and 700 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:32:42 : 3337 steps taken, and 738 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:33:43 : 3446 steps taken, and 770 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:34:43 : 3559 steps taken, and 814 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:35:43 : 3669 steps taken, and 857 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:36:44 : 3779 steps taken, and 892 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:37:44 : 3888 steps taken, and 923 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:38:44 : 3995 steps taken, and 959 ac
cepted.
[mcmc] Learn + convergence test @ 960 samples accepted.
[mcmc]  - Acceptance rate: 0.292
[mcmc]  - Convergence of means: R-1 = 0.383203 after 768 accepted st
eps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 13:39:44 : 4101 steps taken, and 994 ac
cepted.
[mcmc] Progress @ 2026-11-07 13:40:45 : 4212 steps taken, and 1031 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:41:45 : 4324 steps taken, and 1067 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:42:45 : 4441 steps taken, and 1115 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:43:46 : 4551 steps taken, and 1148 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:44:46 : 4663 steps taken, and 1187 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:45:47 : 4774 steps taken, and 1223 a
ccepted.
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[mcmc] Progress @ 2026-11-07 13:46:47 : 4882 steps taken, and 1258 a
ccepted.
[mcmc] Learn + convergence test @ 1280 samples accepted.
[mcmc]  - Acceptance rate: 0.323
[mcmc]  - Convergence of means: R-1 = 0.244032 after 1024 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 13:47:48 : 4990 steps taken, and 1288 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:48:48 : 5096 steps taken, and 1319 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:49:48 : 5205 steps taken, and 1350 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:50:48 : 5319 steps taken, and 1389 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:51:49 : 5431 steps taken, and 1418 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:52:49 : 5529 steps taken, and 1441 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:53:50 : 5618 steps taken, and 1473 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:54:50 : 5711 steps taken, and 1504 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:55:51 : 5801 steps taken, and 1536 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:56:51 : 5893 steps taken, and 1565 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:57:51 : 5981 steps taken, and 1586 a
ccepted.
[mcmc] Learn + convergence test @ 1600 samples accepted.
[mcmc]  - Acceptance rate: 0.342
[mcmc]  - Convergence of means: R-1 = 0.172519 after 1280 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 13:58:52 : 6070 steps taken, and 1616 a
ccepted.
[mcmc] Progress @ 2026-11-07 13:59:52 : 6175 steps taken, and 1653 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:00:52 : 6287 steps taken, and 1693 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:01:52 : 6405 steps taken, and 1728 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:02:53 : 6518 steps taken, and 1763 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:03:53 : 6633 steps taken, and 1795 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:04:54 : 6746 steps taken, and 1826 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:05:54 : 6857 steps taken, and 1851 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:06:55 : 6966 steps taken, and 1887 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:07:55 : 7072 steps taken, and 1911 a
ccepted.
[mcmc] Learn + convergence test @ 1920 samples accepted.
[mcmc]  - Acceptance rate: 0.330
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[mcmc]  - Convergence of means: R-1 = 0.153811 after 1536 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 14:08:55 : 7181 steps taken, and 1947 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:09:56 : 7292 steps taken, and 1980 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:10:56 : 7401 steps taken, and 2017 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:11:56 : 7499 steps taken, and 2042 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:12:57 : 7594 steps taken, and 2075 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:13:57 : 7684 steps taken, and 2093 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:14:58 : 7781 steps taken, and 2118 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:15:58 : 7871 steps taken, and 2144 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:16:58 : 7963 steps taken, and 2173 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:17:58 : 8056 steps taken, and 2198 a
ccepted.
[mcmc] Learn + convergence test @ 2240 samples accepted.
[mcmc]  - Acceptance rate: 0.322
[mcmc]  - Convergence of means: R-1 = 0.101924 after 1792 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 14:18:59 : 8163 steps taken, and 2240 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:19:59 : 8273 steps taken, and 2278 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:21:00 : 8383 steps taken, and 2304 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:22:00 : 8493 steps taken, and 2341 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:23:00 : 8607 steps taken, and 2380 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:24:00 : 8727 steps taken, and 2408 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:25:01 : 8839 steps taken, and 2443 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:26:01 : 8955 steps taken, and 2472 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:27:01 : 9064 steps taken, and 2500 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:28:01 : 9166 steps taken, and 2529 a
ccepted.
[mcmc] Learn + convergence test @ 2560 samples accepted.
[mcmc]  - Acceptance rate: 0.315
[mcmc]  - Convergence of means: R-1 = 0.053698 after 2048 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 14:29:01 : 9275 steps taken, and 2569 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:30:01 : 9384 steps taken, and 2602 a
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ccepted.
[mcmc] Progress @ 2026-11-07 14:31:02 : 9495 steps taken, and 2637 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:32:02 : 9606 steps taken, and 2664 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:33:02 : 9714 steps taken, and 2691 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:34:03 : 9825 steps taken, and 2724 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:35:03 : 9941 steps taken, and 2761 a
ccepted.
[mcmc] Progress @ 2026-11-07 14:36:03 : 10051 steps taken, and 2799
accepted.
[mcmc] Progress @ 2026-11-07 14:37:04 : 10161 steps taken, and 2829
accepted.
[mcmc] Progress @ 2026-11-07 14:38:04 : 10277 steps taken, and 2866
accepted.
[mcmc] Learn + convergence test @ 2880 samples accepted.
[mcmc]  - Acceptance rate: 0.311
[mcmc]  - Convergence of means: R-1 = 0.089696 after 2304 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 14:39:04 : 10389 steps taken, and 2900
accepted.
[mcmc] Progress @ 2026-11-07 14:40:04 : 10502 steps taken, and 2926
accepted.
[mcmc] Progress @ 2026-11-07 14:41:04 : 10618 steps taken, and 2957
accepted.
[mcmc] Progress @ 2026-11-07 14:42:05 : 10728 steps taken, and 2982
accepted.
[mcmc] Progress @ 2026-11-07 14:43:05 : 10840 steps taken, and 3016
accepted.
[mcmc] Progress @ 2026-11-07 14:44:06 : 10952 steps taken, and 3047
accepted.
[mcmc] Progress @ 2026-11-07 14:45:06 : 11072 steps taken, and 3081
accepted.
[mcmc] Progress @ 2026-11-07 14:46:07 : 11188 steps taken, and 3112
accepted.
[mcmc] Progress @ 2026-11-07 14:47:07 : 11298 steps taken, and 3144
accepted.
[mcmc] Progress @ 2026-11-07 14:48:07 : 11405 steps taken, and 3173
accepted.
[mcmc] Learn + convergence test @ 3200 samples accepted.
[mcmc]  - Acceptance rate: 0.303
[mcmc]  - Convergence of means: R-1 = 0.079686 after 2560 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 14:49:07 : 11517 steps taken, and 3209
accepted.
[mcmc] Progress @ 2026-11-07 14:50:07 : 11617 steps taken, and 3243
accepted.
[mcmc] Progress @ 2026-11-07 14:51:08 : 11715 steps taken, and 3268
accepted.
[mcmc] Progress @ 2026-11-07 14:52:08 : 11810 steps taken, and 3298
accepted.
[mcmc] Progress @ 2026-11-07 14:53:08 : 11908 steps taken, and 3335
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accepted.
[mcmc] Progress @ 2026-11-07 14:54:08 : 11998 steps taken, and 3360
accepted.
[mcmc] Progress @ 2026-11-07 14:55:09 : 12101 steps taken, and 3388
accepted.
[mcmc] Progress @ 2026-11-07 14:56:09 : 12201 steps taken, and 3411
accepted.
[mcmc] Progress @ 2026-11-07 14:57:09 : 12282 steps taken, and 3426
accepted.
[mcmc] Progress @ 2026-11-07 14:58:10 : 12376 steps taken, and 3453
accepted.
[mcmc] Progress @ 2026-11-07 14:59:10 : 12473 steps taken, and 3479
accepted.
[mcmc] Progress @ 2026-11-07 15:00:10 : 12569 steps taken, and 3504
accepted.
[mcmc] Learn + convergence test @ 3520 samples accepted.
[mcmc]  - Acceptance rate: 0.300
[mcmc]  - Convergence of means: R-1 = 0.062894 after 2816 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 15:01:10 : 12658 steps taken, and 3529
accepted.
[mcmc] Progress @ 2026-11-07 15:02:11 : 12752 steps taken, and 3557
accepted.
[mcmc] Progress @ 2026-11-07 15:03:11 : 12841 steps taken, and 3588
accepted.
[mcmc] Progress @ 2026-11-07 15:04:12 : 12939 steps taken, and 3624
accepted.
[mcmc] Progress @ 2026-11-07 15:05:13 : 13023 steps taken, and 3653
accepted.
[mcmc] Progress @ 2026-11-07 15:06:13 : 13119 steps taken, and 3683
accepted.
[mcmc] Progress @ 2026-11-07 15:07:13 : 13198 steps taken, and 3703
accepted.
[mcmc] Progress @ 2026-11-07 15:08:13 : 13291 steps taken, and 3725
accepted.
[mcmc] Progress @ 2026-11-07 15:09:14 : 13392 steps taken, and 3750
accepted.
[mcmc] Progress @ 2026-11-07 15:10:14 : 13498 steps taken, and 3776
accepted.
[mcmc] Progress @ 2026-11-07 15:11:15 : 13579 steps taken, and 3793
accepted.
[mcmc] Progress @ 2026-11-07 15:12:15 : 13657 steps taken, and 3812
accepted.
[mcmc] Progress @ 2026-11-07 15:13:16 : 13738 steps taken, and 3833
accepted.
[mcmc] Learn + convergence test @ 3840 samples accepted.
[mcmc]  - Acceptance rate: 0.298
[mcmc]  - Convergence of means: R-1 = 0.065773 after 3072 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 15:14:16 : 13824 steps taken, and 3862
accepted.
[mcmc] Progress @ 2026-11-07 15:15:17 : 13922 steps taken, and 3892
accepted.
[mcmc] Progress @ 2026-11-07 15:16:17 : 14004 steps taken, and 3913
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accepted.
[mcmc] Progress @ 2026-11-07 15:17:17 : 14090 steps taken, and 3938
accepted.
[mcmc] Progress @ 2026-11-07 15:18:17 : 14172 steps taken, and 3960
accepted.
[mcmc] Progress @ 2026-11-07 15:19:17 : 14239 steps taken, and 3980
accepted.
[mcmc] Progress @ 2026-11-07 15:20:18 : 14293 steps taken, and 3997
accepted.
[mcmc] Progress @ 2026-11-07 15:21:18 : 14331 steps taken, and 4008
accepted.
[mcmc] Progress @ 2026-11-07 15:22:20 : 14367 steps taken, and 4023
accepted.
[mcmc] Progress @ 2026-11-07 15:23:21 : 14411 steps taken, and 4037
accepted.
[mcmc] Progress @ 2026-11-07 15:24:21 : 14456 steps taken, and 4051
accepted.
[mcmc] Progress @ 2026-11-07 15:25:23 : 14491 steps taken, and 4061
accepted.
[mcmc] Progress @ 2026-11-07 15:26:23 : 14570 steps taken, and 4081
accepted.
[mcmc] Progress @ 2026-11-07 15:27:23 : 14669 steps taken, and 4114
accepted.
[mcmc] Progress @ 2026-11-07 15:28:23 : 14768 steps taken, and 4144
accepted.
[mcmc] Learn + convergence test @ 4160 samples accepted.
[mcmc]  - Acceptance rate: 0.297
[mcmc]  - Convergence of means: R-1 = 0.037526 after 3328 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 15:29:23 : 14877 steps taken, and 4186
accepted.
[mcmc] Progress @ 2026-11-07 15:30:23 : 14978 steps taken, and 4221
accepted.
[mcmc] Progress @ 2026-11-07 15:31:23 : 15088 steps taken, and 4257
accepted.
[mcmc] Progress @ 2026-11-07 15:32:23 : 15187 steps taken, and 4286
accepted.
[mcmc] Progress @ 2026-11-07 15:33:24 : 15292 steps taken, and 4315
accepted.
[mcmc] Progress @ 2026-11-07 15:34:24 : 15390 steps taken, and 4333
accepted.
[mcmc] Progress @ 2026-11-07 15:35:25 : 15502 steps taken, and 4361
accepted.
[mcmc] Progress @ 2026-11-07 15:36:25 : 15608 steps taken, and 4391
accepted.
[mcmc] Progress @ 2026-11-07 15:37:25 : 15714 steps taken, and 4420
accepted.
[mcmc] Progress @ 2026-11-07 15:38:25 : 15817 steps taken, and 4447
accepted.
[mcmc] Learn + convergence test @ 4480 samples accepted.
[mcmc]  - Acceptance rate: 0.296
[mcmc]  - Convergence of means: R-1 = 0.023422 after 3584 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 15:39:26 : 15923 steps taken, and 4486
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accepted.
[mcmc] Progress @ 2026-11-07 15:40:26 : 16018 steps taken, and 4510
accepted.
[mcmc] Progress @ 2026-11-07 15:41:26 : 16127 steps taken, and 4541
accepted.
[mcmc] Progress @ 2026-11-07 15:42:26 : 16233 steps taken, and 4567
accepted.
[mcmc] Progress @ 2026-11-07 15:43:27 : 16342 steps taken, and 4592
accepted.
[mcmc] Progress @ 2026-11-07 15:44:27 : 16453 steps taken, and 4631
accepted.
[mcmc] Progress @ 2026-11-07 15:45:27 : 16569 steps taken, and 4666
accepted.
[mcmc] Progress @ 2026-11-07 15:46:28 : 16689 steps taken, and 4704
accepted.
[mcmc] Progress @ 2026-11-07 15:47:28 : 16800 steps taken, and 4736
accepted.
[mcmc] Progress @ 2026-11-07 15:48:28 : 16913 steps taken, and 4769
accepted.
[mcmc] Learn + convergence test @ 4800 samples accepted.
[mcmc]  - Acceptance rate: 0.295
[mcmc]  - Convergence of means: R-1 = 0.041024 after 3840 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 15:49:29 : 17024 steps taken, and 4801
accepted.
[mcmc] Progress @ 2026-11-07 15:50:29 : 17134 steps taken, and 4832
accepted.
[mcmc] Progress @ 2026-11-07 15:51:29 : 17241 steps taken, and 4857
accepted.
[mcmc] Progress @ 2026-11-07 15:52:30 : 17352 steps taken, and 4888
accepted.
[mcmc] Progress @ 2026-11-07 15:53:30 : 17460 steps taken, and 4913
accepted.
[mcmc] Progress @ 2026-11-07 15:54:30 : 17572 steps taken, and 4938
accepted.
[mcmc] Progress @ 2026-11-07 15:55:30 : 17680 steps taken, and 4975
accepted.
[mcmc] Progress @ 2026-11-07 15:56:30 : 17789 steps taken, and 5006
accepted.
[mcmc] Progress @ 2026-11-07 15:57:31 : 17899 steps taken, and 5031
accepted.
[mcmc] Progress @ 2026-11-07 15:58:31 : 18010 steps taken, and 5058
accepted.
[mcmc] Progress @ 2026-11-07 15:59:31 : 18119 steps taken, and 5094
accepted.
[mcmc] Learn + convergence test @ 5120 samples accepted.
[mcmc]  - Acceptance rate: 0.292
[mcmc]  - Convergence of means: R-1 = 0.041487 after 4096 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 16:00:32 : 18230 steps taken, and 5127
accepted.
[mcmc] Progress @ 2026-11-07 16:01:32 : 18339 steps taken, and 5152
accepted.
[mcmc] Progress @ 2026-11-07 16:02:32 : 18448 steps taken, and 5181
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accepted.
[mcmc] Progress @ 2026-11-07 16:03:33 : 18562 steps taken, and 5210
accepted.
[mcmc] Progress @ 2026-11-07 16:04:33 : 18675 steps taken, and 5247
accepted.
[mcmc] Progress @ 2026-11-07 16:05:33 : 18781 steps taken, and 5273
accepted.
[mcmc] Progress @ 2026-11-07 16:06:33 : 18891 steps taken, and 5306
accepted.
[mcmc] Progress @ 2026-11-07 16:07:34 : 19005 steps taken, and 5335
accepted.
[mcmc] Progress @ 2026-11-07 16:08:34 : 19113 steps taken, and 5370
accepted.
[mcmc] Progress @ 2026-11-07 16:09:34 : 19221 steps taken, and 5401
accepted.
[mcmc] Progress @ 2026-11-07 16:10:34 : 19326 steps taken, and 5424
accepted.
[mcmc] Learn + convergence test @ 5440 samples accepted.
[mcmc]  - Acceptance rate: 0.290
[mcmc]  - Convergence of means: R-1 = 0.030928 after 4352 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 16:11:35 : 19437 steps taken, and 5450
accepted.
[mcmc] Progress @ 2026-11-07 16:12:35 : 19543 steps taken, and 5477
accepted.
[mcmc] Progress @ 2026-11-07 16:13:35 : 19651 steps taken, and 5511
accepted.
[mcmc] Progress @ 2026-11-07 16:14:35 : 19762 steps taken, and 5533
accepted.
[mcmc] Progress @ 2026-11-07 16:15:36 : 19870 steps taken, and 5571
accepted.
[mcmc] Progress @ 2026-11-07 16:16:36 : 19975 steps taken, and 5604
accepted.
[mcmc] Progress @ 2026-11-07 16:17:36 : 20077 steps taken, and 5629
accepted.
[mcmc] Progress @ 2026-11-07 16:18:36 : 20187 steps taken, and 5659
accepted.
[mcmc] Progress @ 2026-11-07 16:19:36 : 20295 steps taken, and 5683
accepted.
[mcmc] Progress @ 2026-11-07 16:20:36 : 20405 steps taken, and 5712
accepted.
[mcmc] Progress @ 2026-11-07 16:21:37 : 20512 steps taken, and 5741
accepted.
[mcmc] Learn + convergence test @ 5760 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.038563 after 4608 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 16:22:37 : 20618 steps taken, and 5767
accepted.
[mcmc] Progress @ 2026-11-07 16:23:37 : 20726 steps taken, and 5806
accepted.
[mcmc] Progress @ 2026-11-07 16:24:38 : 20836 steps taken, and 5833
accepted.
[mcmc] Progress @ 2026-11-07 16:25:38 : 20949 steps taken, and 5875
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accepted.
[mcmc] Progress @ 2026-11-07 16:26:38 : 21062 steps taken, and 5912
accepted.
[mcmc] Progress @ 2026-11-07 16:27:39 : 21171 steps taken, and 5942
accepted.
[mcmc] Progress @ 2026-11-07 16:28:39 : 21285 steps taken, and 5975
accepted.
[mcmc] Progress @ 2026-11-07 16:29:39 : 21394 steps taken, and 6006
accepted.
[mcmc] Progress @ 2026-11-07 16:30:39 : 21506 steps taken, and 6033
accepted.
[mcmc] Progress @ 2026-11-07 16:31:40 : 21616 steps taken, and 6067
accepted.
[mcmc] Learn + convergence test @ 6080 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.019727 after 4864 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 16:32:40 : 21728 steps taken, and 6103
accepted.
[mcmc] Progress @ 2026-11-07 16:33:40 : 21836 steps taken, and 6125
accepted.
[mcmc] Progress @ 2026-11-07 16:34:41 : 21945 steps taken, and 6157
accepted.
[mcmc] Progress @ 2026-11-07 16:35:41 : 22052 steps taken, and 6190
accepted.
[mcmc] Progress @ 2026-11-07 16:36:41 : 22160 steps taken, and 6216
accepted.
[mcmc] Progress @ 2026-11-07 16:37:41 : 22268 steps taken, and 6243
accepted.
[mcmc] Progress @ 2026-11-07 16:38:42 : 22380 steps taken, and 6279
accepted.
[mcmc] Progress @ 2026-11-07 16:39:42 : 22487 steps taken, and 6300
accepted.
[mcmc] Progress @ 2026-11-07 16:40:42 : 22605 steps taken, and 6343
accepted.
[mcmc] Progress @ 2026-11-07 16:41:43 : 22718 steps taken, and 6376
accepted.
[mcmc] Learn + convergence test @ 6400 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.054470 after 5120 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 16:42:43 : 22823 steps taken, and 6406
accepted.
[mcmc] Progress @ 2026-11-07 16:43:43 : 22937 steps taken, and 6435
accepted.
[mcmc] Progress @ 2026-11-07 16:44:44 : 23048 steps taken, and 6474
accepted.
[mcmc] Progress @ 2026-11-07 16:45:44 : 23159 steps taken, and 6507
accepted.
[mcmc] Progress @ 2026-11-07 16:46:45 : 23268 steps taken, and 6534
accepted.
[mcmc] Progress @ 2026-11-07 16:47:45 : 23377 steps taken, and 6563
accepted.
[mcmc] Progress @ 2026-11-07 16:48:45 : 23481 steps taken, and 6594
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accepted.
[mcmc] Progress @ 2026-11-07 16:49:45 : 23594 steps taken, and 6623
accepted.
[mcmc] Progress @ 2026-11-07 16:50:45 : 23706 steps taken, and 6659
accepted.
[mcmc] Progress @ 2026-11-07 16:51:45 : 23814 steps taken, and 6698
accepted.
[mcmc] Learn + convergence test @ 6720 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.035215 after 5376 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 16:52:45 : 23917 steps taken, and 6728
accepted.
[mcmc] Progress @ 2026-11-07 16:53:46 : 24017 steps taken, and 6761
accepted.
[mcmc] Progress @ 2026-11-07 16:54:46 : 24119 steps taken, and 6785
accepted.
[mcmc] Progress @ 2026-11-07 16:55:46 : 24228 steps taken, and 6817
accepted.
[mcmc] Progress @ 2026-11-07 16:56:46 : 24326 steps taken, and 6843
accepted.
[mcmc] Progress @ 2026-11-07 16:57:46 : 24431 steps taken, and 6871
accepted.
[mcmc] Progress @ 2026-11-07 16:58:47 : 24540 steps taken, and 6899
accepted.
[mcmc] Progress @ 2026-11-07 16:59:47 : 24651 steps taken, and 6938
accepted.
[mcmc] Progress @ 2026-11-07 17:00:47 : 24765 steps taken, and 6970
accepted.
[mcmc] Progress @ 2026-11-07 17:01:47 : 24876 steps taken, and 7001
accepted.
[mcmc] Progress @ 2026-11-07 17:02:48 : 24988 steps taken, and 7037
accepted.
[mcmc] Learn + convergence test @ 7040 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.035809 after 5632 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 17:03:48 : 25101 steps taken, and 7073
accepted.
[mcmc] Progress @ 2026-11-07 17:04:48 : 25216 steps taken, and 7113
accepted.
[mcmc] Progress @ 2026-11-07 17:05:49 : 25329 steps taken, and 7148
accepted.
[mcmc] Progress @ 2026-11-07 17:06:49 : 25442 steps taken, and 7180
accepted.
[mcmc] Progress @ 2026-11-07 17:07:49 : 25553 steps taken, and 7212
accepted.
[mcmc] Progress @ 2026-11-07 17:08:49 : 25664 steps taken, and 7246
accepted.
[mcmc] Progress @ 2026-11-07 17:09:50 : 25782 steps taken, and 7285
accepted.
[mcmc] Progress @ 2026-11-07 17:10:50 : 25891 steps taken, and 7316
accepted.
[mcmc] Progress @ 2026-11-07 17:11:51 : 26008 steps taken, and 7354
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accepted.
[mcmc] Learn + convergence test @ 7360 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.041302 after 5888 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 17:12:51 : 26117 steps taken, and 7376
accepted.
[mcmc] Progress @ 2026-11-07 17:13:52 : 26231 steps taken, and 7410
accepted.
[mcmc] Progress @ 2026-11-07 17:14:52 : 26346 steps taken, and 7444
accepted.
[mcmc] Progress @ 2026-11-07 17:15:53 : 26460 steps taken, and 7477
accepted.
[mcmc] Progress @ 2026-11-07 17:16:53 : 26573 steps taken, and 7504
accepted.
[mcmc] Progress @ 2026-11-07 17:17:53 : 26692 steps taken, and 7535
accepted.
[mcmc] Progress @ 2026-11-07 17:18:53 : 26803 steps taken, and 7567
accepted.
[mcmc] Progress @ 2026-11-07 17:19:53 : 26918 steps taken, and 7603
accepted.
[mcmc] Progress @ 2026-11-07 17:20:53 : 27029 steps taken, and 7637
accepted.
[mcmc] Progress @ 2026-11-07 17:21:53 : 27142 steps taken, and 7671
accepted.
[mcmc] Learn + convergence test @ 7680 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.032531 after 6144 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 17:22:54 : 27251 steps taken, and 7705
accepted.
[mcmc] Progress @ 2026-11-07 17:23:54 : 27361 steps taken, and 7727
accepted.
[mcmc] Progress @ 2026-11-07 17:24:54 : 27477 steps taken, and 7759
accepted.
[mcmc] Progress @ 2026-11-07 17:25:54 : 27586 steps taken, and 7791
accepted.
[mcmc] Progress @ 2026-11-07 17:26:55 : 27699 steps taken, and 7816
accepted.
[mcmc] Progress @ 2026-11-07 17:27:55 : 27809 steps taken, and 7845
accepted.
[mcmc] Progress @ 2026-11-07 17:28:55 : 27918 steps taken, and 7874
accepted.
[mcmc] Progress @ 2026-11-07 17:29:55 : 28033 steps taken, and 7906
accepted.
[mcmc] Progress @ 2026-11-07 17:30:56 : 28142 steps taken, and 7934
accepted.
[mcmc] Progress @ 2026-11-07 17:31:56 : 28251 steps taken, and 7965
accepted.
[mcmc] Progress @ 2026-11-07 17:32:56 : 28365 steps taken, and 8000
accepted.
[mcmc] Learn + convergence test @ 8000 samples accepted.
[mcmc]  - Acceptance rate: 0.286
[mcmc]  - Convergence of means: R-1 = 0.026089 after 6400 accepted s
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teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 17:33:56 : 28474 steps taken, and 8031
accepted.
[mcmc] Progress @ 2026-11-07 17:34:56 : 28587 steps taken, and 8066
accepted.
[mcmc] Progress @ 2026-11-07 17:35:57 : 28684 steps taken, and 8093
accepted.
[mcmc] Progress @ 2026-11-07 17:36:57 : 28782 steps taken, and 8125
accepted.
[mcmc] Progress @ 2026-11-07 17:37:58 : 28879 steps taken, and 8162
accepted.
[mcmc] Progress @ 2026-11-07 17:38:58 : 28981 steps taken, and 8194
accepted.
[mcmc] Progress @ 2026-11-07 17:39:58 : 29101 steps taken, and 8230
accepted.
[mcmc] Progress @ 2026-11-07 17:40:59 : 29214 steps taken, and 8256
accepted.
[mcmc] Progress @ 2026-11-07 17:41:59 : 29324 steps taken, and 8284
accepted.
[mcmc] Learn + convergence test @ 8320 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.020330 after 6656 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 17:42:59 : 29436 steps taken, and 8321
accepted.
[mcmc] Progress @ 2026-11-07 17:43:59 : 29548 steps taken, and 8365
accepted.
[mcmc] Progress @ 2026-11-07 17:45:00 : 29662 steps taken, and 8401
accepted.
[mcmc] Progress @ 2026-11-07 17:46:00 : 29777 steps taken, and 8438
accepted.
[mcmc] Progress @ 2026-11-07 17:47:00 : 29891 steps taken, and 8473
accepted.
[mcmc] Progress @ 2026-11-07 17:48:00 : 30002 steps taken, and 8505
accepted.
[mcmc] Progress @ 2026-11-07 17:49:01 : 30112 steps taken, and 8535
accepted.
[mcmc] Progress @ 2026-11-07 17:50:01 : 30224 steps taken, and 8568
accepted.
[mcmc] Progress @ 2026-11-07 17:51:01 : 30333 steps taken, and 8601
accepted.
[mcmc] Learn + convergence test @ 8640 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.025337 after 6912 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 17:52:02 : 30453 steps taken, and 8641
accepted.
[mcmc] Progress @ 2026-11-07 17:53:02 : 30568 steps taken, and 8676
accepted.
[mcmc] Progress @ 2026-11-07 17:54:03 : 30684 steps taken, and 8715
accepted.
[mcmc] Progress @ 2026-11-07 17:55:03 : 30796 steps taken, and 8741
accepted.
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[mcmc] Progress @ 2026-11-07 17:56:03 : 30906 steps taken, and 8768
accepted.
[mcmc] Progress @ 2026-11-07 17:57:04 : 31020 steps taken, and 8799
accepted.
[mcmc] Progress @ 2026-11-07 17:58:04 : 31130 steps taken, and 8822
accepted.
[mcmc] Progress @ 2026-11-07 17:59:04 : 31245 steps taken, and 8854
accepted.
[mcmc] Progress @ 2026-11-07 18:00:05 : 31355 steps taken, and 8880
accepted.
[mcmc] Progress @ 2026-11-07 18:01:05 : 31471 steps taken, and 8909
accepted.
[mcmc] Progress @ 2026-11-07 18:02:05 : 31585 steps taken, and 8942
accepted.
[mcmc] Learn + convergence test @ 8960 samples accepted.
[mcmc]  - Acceptance rate: 0.286
[mcmc]  - Convergence of means: R-1 = 0.031264 after 7168 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 18:03:05 : 31703 steps taken, and 8973
accepted.
[mcmc] Progress @ 2026-11-07 18:04:06 : 31816 steps taken, and 9006
accepted.
[mcmc] Progress @ 2026-11-07 18:05:06 : 31929 steps taken, and 9037
accepted.
[mcmc] Progress @ 2026-11-07 18:06:06 : 32046 steps taken, and 9077
accepted.
[mcmc] Progress @ 2026-11-07 18:07:06 : 32162 steps taken, and 9109
accepted.
[mcmc] Progress @ 2026-11-07 18:08:07 : 32279 steps taken, and 9142
accepted.
[mcmc] Progress @ 2026-11-07 18:09:07 : 32393 steps taken, and 9176
accepted.
[mcmc] Progress @ 2026-11-07 18:10:07 : 32506 steps taken, and 9208
accepted.
[mcmc] Progress @ 2026-11-07 18:11:07 : 32621 steps taken, and 9245
accepted.
[mcmc] Progress @ 2026-11-07 18:12:08 : 32738 steps taken, and 9276
accepted.
[mcmc] Learn + convergence test @ 9280 samples accepted.
[mcmc]  - Acceptance rate: 0.287
[mcmc]  - Convergence of means: R-1 = 0.036109 after 7424 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 18:13:08 : 32854 steps taken, and 9320
accepted.
[mcmc] Progress @ 2026-11-07 18:14:08 : 32968 steps taken, and 9357
accepted.
[mcmc] Progress @ 2026-11-07 18:15:09 : 33060 steps taken, and 9380
accepted.
[mcmc] Progress @ 2026-11-07 18:16:09 : 33144 steps taken, and 9415
accepted.
[mcmc] Progress @ 2026-11-07 18:17:10 : 33237 steps taken, and 9444
accepted.
[mcmc] Progress @ 2026-11-07 18:18:10 : 33334 steps taken, and 9469
accepted.
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[mcmc] Progress @ 2026-11-07 18:19:11 : 33428 steps taken, and 9498
accepted.
[mcmc] Progress @ 2026-11-07 18:20:11 : 33525 steps taken, and 9531
accepted.
[mcmc] Progress @ 2026-11-07 18:21:11 : 33626 steps taken, and 9560
accepted.
[mcmc] Learn + convergence test @ 9600 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.029112 after 7680 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 18:22:12 : 33756 steps taken, and 9602
accepted.
[mcmc] Progress @ 2026-11-07 18:23:12 : 33859 steps taken, and 9625
accepted.
[mcmc] Progress @ 2026-11-07 18:24:13 : 33963 steps taken, and 9661
accepted.
[mcmc] Progress @ 2026-11-07 18:25:13 : 34080 steps taken, and 9694
accepted.
[mcmc] Progress @ 2026-11-07 18:26:13 : 34192 steps taken, and 9729
accepted.
[mcmc] Progress @ 2026-11-07 18:27:14 : 34295 steps taken, and 9760
accepted.
[mcmc] Progress @ 2026-11-07 18:28:14 : 34406 steps taken, and 9792
accepted.
[mcmc] Progress @ 2026-11-07 18:29:15 : 34507 steps taken, and 9832
accepted.
[mcmc] Progress @ 2026-11-07 18:30:15 : 34600 steps taken, and 9861
accepted.
[mcmc] Progress @ 2026-11-07 18:31:16 : 34710 steps taken, and 9895
accepted.
[mcmc] Learn + convergence test @ 9920 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.028963 after 7936 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 18:32:16 : 34813 steps taken, and 9925
accepted.
[mcmc] Progress @ 2026-11-07 18:33:16 : 34909 steps taken, and 9953
accepted.
[mcmc] Progress @ 2026-11-07 18:34:16 : 35000 steps taken, and 9975
accepted.
[mcmc] Progress @ 2026-11-07 18:35:17 : 35094 steps taken, and 10001
accepted.
[mcmc] Progress @ 2026-11-07 18:36:17 : 35177 steps taken, and 10030
accepted.
[mcmc] Progress @ 2026-11-07 18:37:18 : 35271 steps taken, and 10057
accepted.
[mcmc] Progress @ 2026-11-07 18:38:18 : 35357 steps taken, and 10081
accepted.
[mcmc] Progress @ 2026-11-07 18:39:19 : 35452 steps taken, and 10114
accepted.
[mcmc] Progress @ 2026-11-07 18:40:19 : 35548 steps taken, and 10138
accepted.
[mcmc] Progress @ 2026-11-07 18:41:20 : 35650 steps taken, and 10165
accepted.
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[mcmc] Progress @ 2026-11-07 18:42:20 : 35741 steps taken, and 10190
accepted.
[mcmc] Progress @ 2026-11-07 18:43:20 : 35833 steps taken, and 10220
accepted.
[mcmc] Learn + convergence test @ 10240 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.017998 after 8192 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 18:44:21 : 35920 steps taken, and 10249
accepted.
[mcmc] Progress @ 2026-11-07 18:45:21 : 36007 steps taken, and 10263
accepted.
[mcmc] Progress @ 2026-11-07 18:46:22 : 36103 steps taken, and 10295
accepted.
[mcmc] Progress @ 2026-11-07 18:47:22 : 36185 steps taken, and 10316
accepted.
[mcmc] Progress @ 2026-11-07 18:48:22 : 36272 steps taken, and 10335
accepted.
[mcmc] Progress @ 2026-11-07 18:49:22 : 36350 steps taken, and 10358
accepted.
[mcmc] Progress @ 2026-11-07 18:50:23 : 36440 steps taken, and 10383
accepted.
[mcmc] Progress @ 2026-11-07 18:51:23 : 36531 steps taken, and 10403
accepted.
[mcmc] Progress @ 2026-11-07 18:52:24 : 36618 steps taken, and 10427
accepted.
[mcmc] Progress @ 2026-11-07 18:53:24 : 36690 steps taken, and 10452
accepted.
[mcmc] Progress @ 2026-11-07 18:54:25 : 36757 steps taken, and 10475
accepted.
[mcmc] Progress @ 2026-11-07 18:55:25 : 36819 steps taken, and 10490
accepted.
[mcmc] Progress @ 2026-11-07 18:56:26 : 36896 steps taken, and 10516
accepted.
[mcmc] Progress @ 2026-11-07 18:57:26 : 37001 steps taken, and 10548
accepted.
[mcmc] Learn + convergence test @ 10560 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.014705 after 8448 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 18:58:26 : 37087 steps taken, and 10564
accepted.
[mcmc] Progress @ 2026-11-07 18:59:26 : 37175 steps taken, and 10589
accepted.
[mcmc] Progress @ 2026-11-07 19:00:26 : 37275 steps taken, and 10621
accepted.
[mcmc] Progress @ 2026-11-07 19:01:26 : 37381 steps taken, and 10666
accepted.
[mcmc] Progress @ 2026-11-07 19:02:27 : 37486 steps taken, and 10702
accepted.
[mcmc] Progress @ 2026-11-07 19:03:27 : 37595 steps taken, and 10727
accepted.
[mcmc] Progress @ 2026-11-07 19:04:27 : 37702 steps taken, and 10757
accepted.
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[mcmc] Progress @ 2026-11-07 19:05:28 : 37806 steps taken, and 10783
accepted.
[mcmc] Progress @ 2026-11-07 19:06:28 : 37914 steps taken, and 10813
accepted.
[mcmc] Progress @ 2026-11-07 19:07:29 : 38020 steps taken, and 10845
accepted.
[mcmc] Progress @ 2026-11-07 19:08:29 : 38119 steps taken, and 10874
accepted.
[mcmc] Learn + convergence test @ 10880 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.023057 after 8704 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 19:09:29 : 38200 steps taken, and 10901
accepted.
[mcmc] Progress @ 2026-11-07 19:10:30 : 38304 steps taken, and 10931
accepted.
[mcmc] Progress @ 2026-11-07 19:11:30 : 38417 steps taken, and 10958
accepted.
[mcmc] Progress @ 2026-11-07 19:12:30 : 38531 steps taken, and 10997
accepted.
[mcmc] Progress @ 2026-11-07 19:13:30 : 38640 steps taken, and 11026
accepted.
[mcmc] Progress @ 2026-11-07 19:14:30 : 38752 steps taken, and 11059
accepted.
[mcmc] Progress @ 2026-11-07 19:15:30 : 38865 steps taken, and 11095
accepted.
[mcmc] Progress @ 2026-11-07 19:16:31 : 38976 steps taken, and 11141
accepted.
[mcmc] Progress @ 2026-11-07 19:17:31 : 39091 steps taken, and 11182
accepted.
[mcmc] Learn + convergence test @ 11200 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.020627 after 8960 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 19:18:31 : 39199 steps taken, and 11212
accepted.
[mcmc] Progress @ 2026-11-07 19:19:31 : 39310 steps taken, and 11245
accepted.
[mcmc] Progress @ 2026-11-07 19:20:31 : 39418 steps taken, and 11274
accepted.
[mcmc] Progress @ 2026-11-07 19:21:32 : 39527 steps taken, and 11302
accepted.
[mcmc] Progress @ 2026-11-07 19:22:32 : 39640 steps taken, and 11333
accepted.
[mcmc] Progress @ 2026-11-07 19:23:33 : 39760 steps taken, and 11376
accepted.
[mcmc] Progress @ 2026-11-07 19:24:33 : 39870 steps taken, and 11404
accepted.
[mcmc] Progress @ 2026-11-07 19:25:33 : 39980 steps taken, and 11429
accepted.
[mcmc] Progress @ 2026-11-07 19:26:33 : 40087 steps taken, and 11454
accepted.
[mcmc] Progress @ 2026-11-07 19:27:34 : 40198 steps taken, and 11483
accepted.
```

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[mcmc] Learn + convergence test @ 11520 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.026361 after 9216 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 19:28:34 : 40309 steps taken, and 11521
accepted.
[mcmc] Progress @ 2026-11-07 19:29:35 : 40400 steps taken, and 11550
accepted.
[mcmc] Progress @ 2026-11-07 19:30:35 : 40496 steps taken, and 11572
accepted.
[mcmc] Progress @ 2026-11-07 19:31:35 : 40606 steps taken, and 11599
accepted.
[mcmc] Progress @ 2026-11-07 19:32:35 : 40724 steps taken, and 11630
accepted.
[mcmc] Progress @ 2026-11-07 19:33:35 : 40832 steps taken, and 11656
accepted.
[mcmc] Progress @ 2026-11-07 19:34:35 : 40944 steps taken, and 11691
accepted.
[mcmc] Progress @ 2026-11-07 19:35:36 : 41060 steps taken, and 11725
accepted.
[mcmc] Progress @ 2026-11-07 19:36:37 : 41172 steps taken, and 11757
accepted.
[mcmc] Progress @ 2026-11-07 19:37:37 : 41285 steps taken, and 11792
accepted.
[mcmc] Progress @ 2026-11-07 19:38:37 : 41397 steps taken, and 11829
accepted.
[mcmc] Learn + convergence test @ 11840 samples accepted.
[mcmc]  - Acceptance rate: 0.288
[mcmc]  - Convergence of means: R-1 = 0.025988 after 9472 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 19:39:38 : 41506 steps taken, and 11861
accepted.
[mcmc] Progress @ 2026-11-07 19:40:38 : 41620 steps taken, and 11896
accepted.
[mcmc] Progress @ 2026-11-07 19:41:38 : 41728 steps taken, and 11927
accepted.
[mcmc] Progress @ 2026-11-07 19:42:38 : 41847 steps taken, and 11952
accepted.
[mcmc] Progress @ 2026-11-07 19:43:38 : 41954 steps taken, and 11990
accepted.
[mcmc] Progress @ 2026-11-07 19:44:39 : 42066 steps taken, and 12027
accepted.
[mcmc] Progress @ 2026-11-07 19:45:39 : 42179 steps taken, and 12068
accepted.
[mcmc] Progress @ 2026-11-07 19:46:40 : 42293 steps taken, and 12099
accepted.
[mcmc] Progress @ 2026-11-07 19:47:40 : 42404 steps taken, and 12131
accepted.
[mcmc] Learn + convergence test @ 12160 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.022893 after 9728 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 19:48:40 : 42518 steps taken, and 12160
```

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accepted.
[mcmc] Progress @ 2026-11-07 19:49:40 : 42631 steps taken, and 12190
accepted.
[mcmc] Progress @ 2026-11-07 19:50:40 : 42741 steps taken, and 12218
accepted.
[mcmc] Progress @ 2026-11-07 19:51:41 : 42852 steps taken, and 12252
accepted.
[mcmc] Progress @ 2026-11-07 19:52:41 : 42959 steps taken, and 12266
accepted.
[mcmc] Progress @ 2026-11-07 19:53:41 : 43070 steps taken, and 12297
accepted.
[mcmc] Progress @ 2026-11-07 19:54:42 : 43185 steps taken, and 12329
accepted.
[mcmc] Progress @ 2026-11-07 19:55:42 : 43296 steps taken, and 12366
accepted.
[mcmc] Progress @ 2026-11-07 19:56:42 : 43403 steps taken, and 12393
accepted.
[mcmc] Progress @ 2026-11-07 19:57:43 : 43515 steps taken, and 12431
accepted.
[mcmc] Progress @ 2026-11-07 19:58:43 : 43636 steps taken, and 12474
accepted.
[mcmc] Learn + convergence test @ 12480 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.018999 after 9984 accepted s
teps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 19:59:44 : 43749 steps taken, and 12514
accepted.
[mcmc] Progress @ 2026-11-07 20:00:44 : 43856 steps taken, and 12546
accepted.
[mcmc] Progress @ 2026-11-07 20:01:44 : 43966 steps taken, and 12580
accepted.
[mcmc] Progress @ 2026-11-07 20:02:44 : 44076 steps taken, and 12611
accepted.
[mcmc] Progress @ 2026-11-07 20:03:45 : 44189 steps taken, and 12649
accepted.
[mcmc] Progress @ 2026-11-07 20:04:45 : 44307 steps taken, and 12680
accepted.
[mcmc] Progress @ 2026-11-07 20:05:45 : 44423 steps taken, and 12717
accepted.
[mcmc] Progress @ 2026-11-07 20:06:46 : 44536 steps taken, and 12753
accepted.
[mcmc] Progress @ 2026-11-07 20:07:46 : 44653 steps taken, and 12784
accepted.
[mcmc] Learn + convergence test @ 12800 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.021721 after 10240 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 20:08:47 : 44767 steps taken, and 12823
accepted.
[mcmc] Progress @ 2026-11-07 20:09:47 : 44880 steps taken, and 12852
accepted.
[mcmc] Progress @ 2026-11-07 20:10:47 : 44991 steps taken, and 12879
accepted.
[mcmc] Progress @ 2026-11-07 20:11:48 : 45105 steps taken, and 12919
```

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accepted.
[mcmc] Progress @ 2026-11-07 20:12:48 : 45216 steps taken, and 12948
accepted.
[mcmc] Progress @ 2026-11-07 20:13:49 : 45327 steps taken, and 12979
accepted.
[mcmc] Progress @ 2026-11-07 20:14:49 : 45439 steps taken, and 13011
accepted.
[mcmc] Progress @ 2026-11-07 20:15:50 : 45554 steps taken, and 13051
accepted.
[mcmc] Progress @ 2026-11-07 20:16:50 : 45666 steps taken, and 13086
accepted.
[mcmc] Progress @ 2026-11-07 20:17:50 : 45783 steps taken, and 13116
accepted.
[mcmc] Learn + convergence test @ 13120 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.022496 after 10496 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 20:18:50 : 45894 steps taken, and 13150
accepted.
[mcmc] Progress @ 2026-11-07 20:19:51 : 46004 steps taken, and 13184
accepted.
[mcmc] Progress @ 2026-11-07 20:20:51 : 46117 steps taken, and 13225
accepted.
[mcmc] Progress @ 2026-11-07 20:21:51 : 46230 steps taken, and 13254
accepted.
[mcmc] Progress @ 2026-11-07 20:22:51 : 46343 steps taken, and 13289
accepted.
[mcmc] Progress @ 2026-11-07 20:23:52 : 46458 steps taken, and 13330
accepted.
[mcmc] Progress @ 2026-11-07 20:24:52 : 46569 steps taken, and 13372
accepted.
[mcmc] Progress @ 2026-11-07 20:25:52 : 46680 steps taken, and 13396
accepted.
[mcmc] Progress @ 2026-11-07 20:26:52 : 46796 steps taken, and 13430
accepted.
[mcmc] Learn + convergence test @ 13440 samples accepted.
[mcmc]  - Acceptance rate: 0.290
[mcmc]  - Convergence of means: R-1 = 0.023837 after 10752 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 20:27:52 : 46907 steps taken, and 13469
accepted.
[mcmc] Progress @ 2026-11-07 20:28:53 : 47020 steps taken, and 13507
accepted.
[mcmc] Progress @ 2026-11-07 20:29:53 : 47130 steps taken, and 13539
accepted.
[mcmc] Progress @ 2026-11-07 20:30:53 : 47246 steps taken, and 13578
accepted.
[mcmc] Progress @ 2026-11-07 20:31:54 : 47370 steps taken, and 13619
accepted.
[mcmc] Progress @ 2026-11-07 20:32:54 : 47483 steps taken, and 13657
accepted.
[mcmc] Progress @ 2026-11-07 20:33:55 : 47598 steps taken, and 13685
accepted.
[mcmc] Progress @ 2026-11-07 20:34:55 : 47714 steps taken, and 13717
```

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accepted.
[mcmc] Progress @ 2026-11-07 20:35:55 : 47831 steps taken, and 13756
accepted.
[mcmc] Learn + convergence test @ 13760 samples accepted.
[mcmc]  - Acceptance rate: 0.290
[mcmc]  - Convergence of means: R-1 = 0.020902 after 11008 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 20:36:56 : 47945 steps taken, and 13795
accepted.
[mcmc] Progress @ 2026-11-07 20:37:56 : 48064 steps taken, and 13822
accepted.
[mcmc] Progress @ 2026-11-07 20:38:57 : 48178 steps taken, and 13859
accepted.
[mcmc] Progress @ 2026-11-07 20:39:57 : 48292 steps taken, and 13895
accepted.
[mcmc] Progress @ 2026-11-07 20:40:57 : 48408 steps taken, and 13928
accepted.
[mcmc] Progress @ 2026-11-07 20:41:57 : 48519 steps taken, and 13956
accepted.
[mcmc] Progress @ 2026-11-07 20:42:57 : 48632 steps taken, and 13990
accepted.
[mcmc] Progress @ 2026-11-07 20:43:58 : 48748 steps taken, and 14023
accepted.
[mcmc] Progress @ 2026-11-07 20:44:58 : 48864 steps taken, and 14063
accepted.
[mcmc] Learn + convergence test @ 14080 samples accepted.
[mcmc]  - Acceptance rate: 0.290
[mcmc]  - Convergence of means: R-1 = 0.016730 after 11264 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 20:45:59 : 48974 steps taken, and 14083
accepted.
[mcmc] Progress @ 2026-11-07 20:46:59 : 49082 steps taken, and 14107
accepted.
[mcmc] Progress @ 2026-11-07 20:47:59 : 49190 steps taken, and 14133
accepted.
[mcmc] Progress @ 2026-11-07 20:48:59 : 49298 steps taken, and 14152
accepted.
[mcmc] Progress @ 2026-11-07 20:50:00 : 49418 steps taken, and 14187
accepted.
[mcmc] Progress @ 2026-11-07 20:51:00 : 49528 steps taken, and 14216
accepted.
[mcmc] Progress @ 2026-11-07 20:52:00 : 49640 steps taken, and 14242
accepted.
[mcmc] Progress @ 2026-11-07 20:53:01 : 49756 steps taken, and 14268
accepted.
[mcmc] Progress @ 2026-11-07 20:54:01 : 49867 steps taken, and 14300
accepted.
[mcmc] Progress @ 2026-11-07 20:55:01 : 49981 steps taken, and 14331
accepted.
[mcmc] Progress @ 2026-11-07 20:56:02 : 50094 steps taken, and 14363
accepted.
[mcmc] Learn + convergence test @ 14400 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.020702 after 11520 accepted
```

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steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 20:57:02 : 50202 steps taken, and 14401
accepted.
[mcmc] Progress @ 2026-11-07 20:58:03 : 50314 steps taken, and 14441
accepted.
[mcmc] Progress @ 2026-11-07 20:59:03 : 50410 steps taken, and 14460
accepted.
[mcmc] Progress @ 2026-11-07 21:00:03 : 50512 steps taken, and 14489
accepted.
[mcmc] Progress @ 2026-11-07 21:01:03 : 50604 steps taken, and 14515
accepted.
[mcmc] Progress @ 2026-11-07 21:02:04 : 50697 steps taken, and 14539
accepted.
[mcmc] Progress @ 2026-11-07 21:03:04 : 50797 steps taken, and 14569
accepted.
[mcmc] Progress @ 2026-11-07 21:04:05 : 50907 steps taken, and 14596
accepted.
[mcmc] Progress @ 2026-11-07 21:05:05 : 51014 steps taken, and 14628
accepted.
[mcmc] Progress @ 2026-11-07 21:06:06 : 51111 steps taken, and 14660
accepted.
[mcmc] Progress @ 2026-11-07 21:07:06 : 51206 steps taken, and 14692
accepted.
[mcmc] Learn + convergence test @ 14720 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.023788 after 11776 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 21:08:06 : 51312 steps taken, and 14726
accepted.
[mcmc] Progress @ 2026-11-07 21:09:06 : 51414 steps taken, and 14754
accepted.
[mcmc] Progress @ 2026-11-07 21:10:06 : 51517 steps taken, and 14780
accepted.
[mcmc] Progress @ 2026-11-07 21:11:06 : 51625 steps taken, and 14815
accepted.
[mcmc] Progress @ 2026-11-07 21:12:07 : 51734 steps taken, and 14833
accepted.
[mcmc] Progress @ 2026-11-07 21:13:07 : 51845 steps taken, and 14878
accepted.
[mcmc] Progress @ 2026-11-07 21:14:07 : 51955 steps taken, and 14905
accepted.
[mcmc] Progress @ 2026-11-07 21:15:07 : 52072 steps taken, and 14927
accepted.
[mcmc] Progress @ 2026-11-07 21:16:08 : 52184 steps taken, and 14954
accepted.
[mcmc] Progress @ 2026-11-07 21:17:08 : 52294 steps taken, and 14988
accepted.
[mcmc] Progress @ 2026-11-07 21:18:08 : 52403 steps taken, and 15022
accepted.
[mcmc] Learn + convergence test @ 15040 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.017046 after 12032 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
```

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[mcmc] Progress @ 2026-11-07 21:19:09 : 52515 steps taken, and 15060
accepted.
[mcmc] Progress @ 2026-11-07 21:20:09 : 52629 steps taken, and 15093
accepted.
[mcmc] Progress @ 2026-11-07 21:21:09 : 52743 steps taken, and 15124
accepted.
[mcmc] Progress @ 2026-11-07 21:22:09 : 52851 steps taken, and 15161
accepted.
[mcmc] Progress @ 2026-11-07 21:23:09 : 52961 steps taken, and 15192
accepted.
[mcmc] Progress @ 2026-11-07 21:24:09 : 53070 steps taken, and 15222
accepted.
[mcmc] Progress @ 2026-11-07 21:25:10 : 53180 steps taken, and 15253
accepted.
[mcmc] Progress @ 2026-11-07 21:26:10 : 53285 steps taken, and 15283
accepted.
[mcmc] Progress @ 2026-11-07 21:27:10 : 53397 steps taken, and 15317
accepted.
[mcmc] Progress @ 2026-11-07 21:28:11 : 53508 steps taken, and 15354
accepted.
[mcmc] Learn + convergence test @ 15360 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.014451 after 12288 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 21:29:11 : 53619 steps taken, and 15392
accepted.
[mcmc] Progress @ 2026-11-07 21:30:11 : 53726 steps taken, and 15421
accepted.
[mcmc] Progress @ 2026-11-07 21:31:12 : 53832 steps taken, and 15448
accepted.
[mcmc] Progress @ 2026-11-07 21:32:12 : 53938 steps taken, and 15467
accepted.
[mcmc] Progress @ 2026-11-07 21:33:12 : 54048 steps taken, and 15498
accepted.
[mcmc] Progress @ 2026-11-07 21:34:13 : 54159 steps taken, and 15527
accepted.
[mcmc] Progress @ 2026-11-07 21:35:13 : 54273 steps taken, and 15567
accepted.
[mcmc] Progress @ 2026-11-07 21:36:13 : 54388 steps taken, and 15603
accepted.
[mcmc] Progress @ 2026-11-07 21:37:13 : 54501 steps taken, and 15637
accepted.
[mcmc] Progress @ 2026-11-07 21:38:14 : 54609 steps taken, and 15665
accepted.
[mcmc] Learn + convergence test @ 15680 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.010813 after 12544 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 21:39:14 : 54726 steps taken, and 15694
accepted.
[mcmc] Progress @ 2026-11-07 21:40:14 : 54835 steps taken, and 15724
accepted.
[mcmc] Progress @ 2026-11-07 21:41:15 : 54956 steps taken, and 15762
accepted.
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[mcmc] Progress @ 2026-11-07 21:42:15 : 55066 steps taken, and 15799
accepted.
[mcmc] Progress @ 2026-11-07 21:43:16 : 55182 steps taken, and 15839
accepted.
[mcmc] Progress @ 2026-11-07 21:44:16 : 55292 steps taken, and 15875
accepted.
[mcmc] Progress @ 2026-11-07 21:45:17 : 55405 steps taken, and 15915
accepted.
[mcmc] Progress @ 2026-11-07 21:46:17 : 55512 steps taken, and 15941
accepted.
[mcmc] Progress @ 2026-11-07 21:47:17 : 55617 steps taken, and 15973
accepted.
[mcmc] Progress @ 2026-11-07 21:48:18 : 55728 steps taken, and 15997
accepted.
[mcmc] Learn + convergence test @ 16000 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.011985 after 12800 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 21:49:18 : 55840 steps taken, and 16027
accepted.
[mcmc] Progress @ 2026-11-07 21:50:18 : 55951 steps taken, and 16055
accepted.
[mcmc] Progress @ 2026-11-07 21:51:19 : 56060 steps taken, and 16091
accepted.
[mcmc] Progress @ 2026-11-07 21:52:19 : 56171 steps taken, and 16125
accepted.
[mcmc] Progress @ 2026-11-07 21:53:19 : 56284 steps taken, and 16157
accepted.
[mcmc] Progress @ 2026-11-07 21:54:20 : 56396 steps taken, and 16188
accepted.
[mcmc] Progress @ 2026-11-07 21:55:20 : 56506 steps taken, and 16224
accepted.
[mcmc] Progress @ 2026-11-07 21:56:21 : 56622 steps taken, and 16261
accepted.
[mcmc] Progress @ 2026-11-07 21:57:21 : 56733 steps taken, and 16298
accepted.
[mcmc] Learn + convergence test @ 16320 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.009209 after 13056 accepted
steps
[mcmc]  - Updated covariance matrix of proposal pdf.
[mcmc] Progress @ 2026-11-07 21:58:21 : 56839 steps taken, and 16332
accepted.
[mcmc] Progress @ 2026-11-07 21:59:21 : 56943 steps taken, and 16356
accepted.
[mcmc] Progress @ 2026-11-07 22:00:22 : 57052 steps taken, and 16387
accepted.
[mcmc] Progress @ 2026-11-07 22:01:22 : 57160 steps taken, and 16420
accepted.
[mcmc] Progress @ 2026-11-07 22:02:22 : 57268 steps taken, and 16452
accepted.
[mcmc] Progress @ 2026-11-07 22:03:22 : 57381 steps taken, and 16490
accepted.
[mcmc] Progress @ 2026-11-07 22:04:22 : 57493 steps taken, and 16520
accepted.
```

```
[mcmc] Progress @ 2026-11-07 22:05:22 : 57610 steps taken, and 16553
accepted.
[mcmc] Progress @ 2026-11-07 22:06:23 : 57724 steps taken, and 16585
accepted.
[mcmc] Progress @ 2026-11-07 22:07:23 : 57833 steps taken, and 16615
accepted.
[mcmc] Learn + convergence test @ 16640 samples accepted.
[mcmc]  - Acceptance rate: 0.289
[mcmc]  - Convergence of means: R-1 = 0.009641 after 13312 accepted
steps
[mcmc]  - Convergence of bounds: R-1 = 0.063171 after 16640 accepted
steps
[mcmc] The run has converged!
[mcmc] Sampling complete after 16640 accepted steps.
Done.
```

Cell 9b : Load the saved chain

In [13]:
```
from getdist import loadMCSamples, plots, MCSamples

CHAIN_ROOT = CHAIN_DIR + f'/{TAG}_w0wa'
gd = loadMCSamples(CHAIN_ROOT, settings={'ignore_rows': 0.3})   # f
print(f'Loaded {CHAIN_ROOT}: {gd.numrows} weighted samples')
```

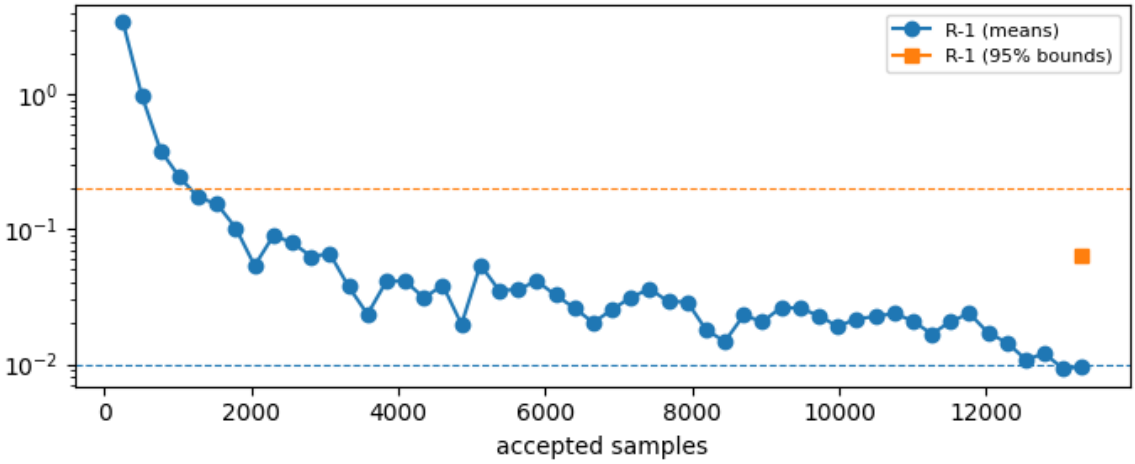
```
Loaded /Users/cherianpi/Desktop/Cosmology/analyses/wmap9_desi_dr2_pa
ntheon_w0wa/chains/wmap9_dr2_pantheonplus_w0wa: 11648 weighted sampl
es
```

Cell 10 : Convergence Diagnostics

In [14]:
```
prog = np.atleast_1d(np.genfromtxt(CHAIN_ROOT + '.progress', names=
if prog.size == 0 or not np.isfinite(prog['N']).any():
    print('No convergence checkpoints written yet (the chain is sti
else:
    print('Cobaya progress (last 5 checkpoints):')
    print(f'{"N accepted":>12} {"acceptance":>11} {"R-1 (means)":>1
    for row in prog[-5:]:
        print(f'{int(row["N"]):12d} {row["acceptance_rate"]:11.3f}

    fig, ax = plt.subplots(figsize=(7, 3))
    ax.semilogy(prog['N'], prog['Rminus1'], 'o-', label='R-1 (means
    ax.semilogy(prog['N'], prog['Rminus1_cl'], 's-', label='R-1 (95
    ax.axhline(0.01, color='C0', ls='--', lw=0.8); ax.axhline(0.2,
    ax.set_xlabel('accepted samples'); ax.legend(fontsize=8)
    fig.tight_layout(); fig.savefig(IMG_DIR + f'/{TAG}_convergence.
```

```
Cobaya progress (last 5 checkpoints):
  N accepted  acceptance  R-1 (means)  R-1 (95%)
       12288       0.289       0.0145        nan
       12544       0.289       0.0108        nan
       12800       0.289       0.0120        nan
       13056       0.289       0.0092        nan
       13312       0.289       0.0096     0.0632
```



Cell 11 : Posterior Summary

Includes

1. the derived Ω_m.
2. the **pivot** equation of state: $a_p = 1 + \mathrm{Cov}(w_0, w_a)/\mathrm{Var}(w_a)$,

$w_p = w_0 + w_a(1 - a_p)$, 3. w_0–w_a contour.

In [15]:
```
print(f'=== {DATA_LABEL} — w0waCDM Posterior ===')
for p in PARAM_NAMES + ['omegam', 'rdrag']:
    print('  ' + gd.getInlineLatex(p, limit=1))

w0_s = gd.samples[:, gd.index['w']]
wa_s = gd.samples[:, gd.index['wa']]
wts  = gd.weights
cov_w = np.cov(np.vstack([w0_s, wa_s]), aweights=wts)
a_p   = 1.0 + cov_w[0, 1] / cov_w[1, 1]
wp_s  = w0_s + wa_s * (1.0 - a_p)
wp_mean = np.average(wp_s, weights=wts)
wp_std  = np.sqrt(np.average((wp_s - wp_mean)**2, weights=wts))
print(f'\nPivot: z_p = {1/a_p - 1:.3f},  w_p = {wp_mean:.3f} ± {wp_

mu  = np.array([np.average(w0_s, weights=wts), np.average(wa_s, wei
d   = mu - np.array([-1.0, 0.0])
c2  = float(d @ np.linalg.solve(cov_w, d))
sig = stats.norm.isf(stats.chi2.sf(c2, 2) / 2)
print(f'LCDM point from the w0–wa posterior: {sig:.2f} sigma (Gauss
```

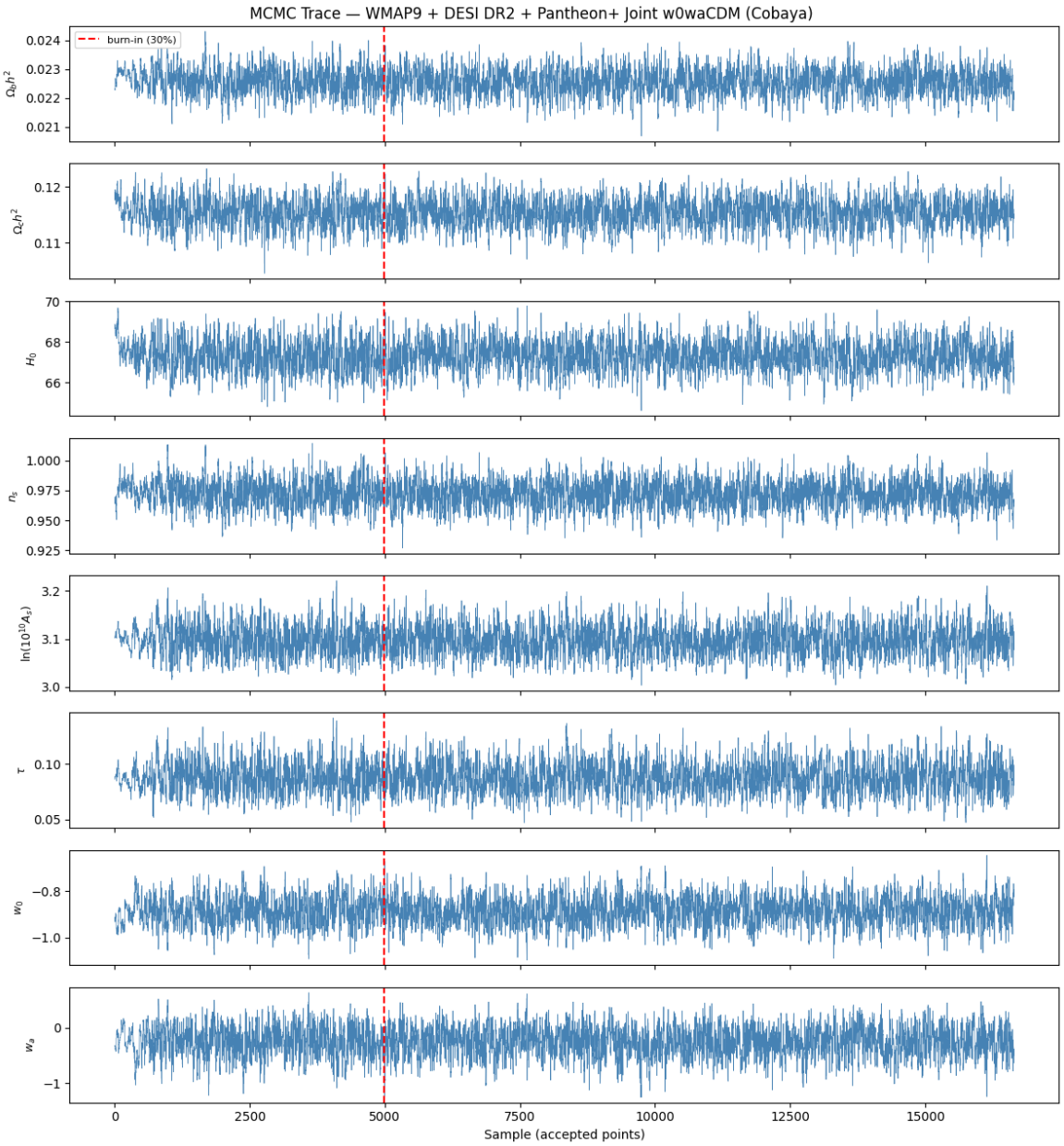
```
=== WMAP9 + DESI DR2 + Pantheon+ — w0waCDM Posterior ===
  \Omega_b h^2 = 0.02255\pm 0.00043
  \Omega_c h^2 = 0.1153\pm 0.0024
  H_0 = 67.33\pm 0.70
  n_s = 0.972\pm 0.010
  \ln(10^{10}A_s) = 3.096\pm 0.029
  \tau = 0.087^{+0.012}_{-0.014}
  w_0 = -0.886\pm 0.059
  w_a = -0.25^{+0.28}_{-0.24}
  \Omega_m = 0.3056\pm 0.0063
  r_d = 148.15\pm 0.87

Pivot: z_p = 0.247,  w_p = -0.936 ± 0.028
LCDM point from the w0-wa posterior: 2.01 sigma (Gaussian approximat
ion)
```

Cell 12 : Trace Plot

In [16]:

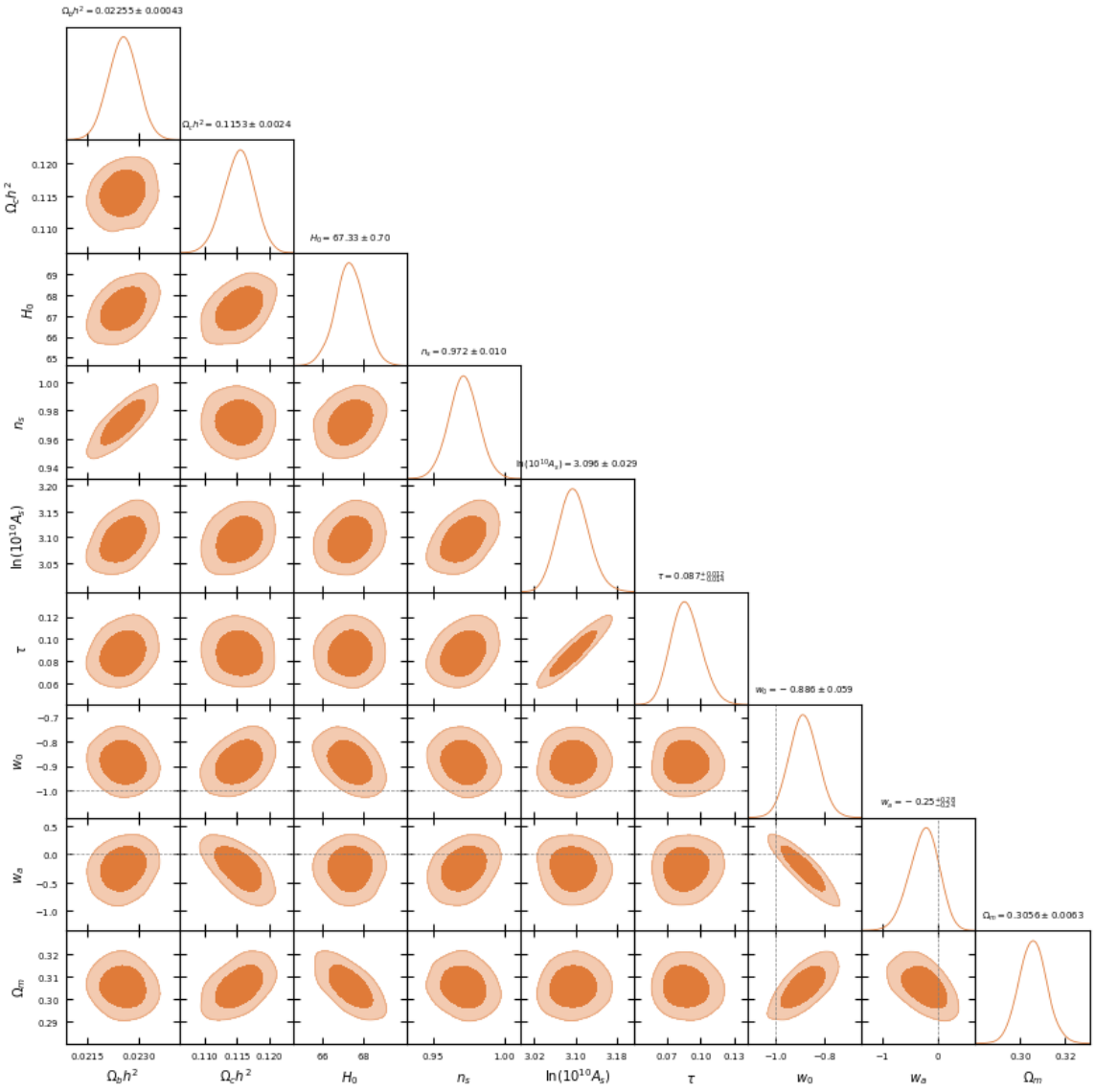
```
with open(CHAIN_ROOT + '.1.txt') as fh:                    # Cobaya ch
    cols = fh.readline().lstrip('#').split()
    raw  = np.loadtxt(fh)
burn = int(0.3 * len(raw))
fig, axes = plt.subplots(len(PARAM_NAMES), 1, figsize=(12, 13), sha
for ax, name, lbl in zip(axes, PARAM_NAMES, PARAM_LABELS):
    ax.plot(raw[:, cols.index(name)], lw=0.5, color='steelblue')
    ax.axvline(burn, color='red', ls='--', lw=1.5, label='burn-in (
    ax.set_ylabel(lbl, fontsize=9)
axes[0].legend(fontsize=8)
axes[-1].set_xlabel('Sample (accepted points)')
fig.suptitle(f'MCMC Trace — {DATA_LABEL} Joint w0waCDM (Cobaya)', f
plt.tight_layout()
fname = IMG_DIR + f'/{TAG}_trace.png'
plt.savefig(fname, dpi=150, bbox_inches='tight')
plt.show()
print(f'Saved: {fname}')
```




```
Saved: /Users/cherianpi/Desktop/Cosmology/analyses/wmap9_desi_dr2_pa
ntheon_w0wa/figures/wmap9_dr2_pantheonplus_trace.png
```

Cell 13 : Corner Plot

In [17]:
```
g = plots.get_subplot_plotter(width_inch=10)
g.settings.axes_fontsize = 9
g.settings.axes_labelsize = 12
g.settings.title_limit_fontsize = 9
g.triangle_plot([gd], PARAM_NAMES + ['omegam'], filled=True,
                contour_colors=['#E07B39'], title_limit=1,
                markers={'w': -1.0, 'wa': 0.0})            # LCDM v
g.export(IMG_DIR + f'/{TAG}_corner.pdf')
g.export(IMG_DIR + f'/{TAG}_corner.png')
plt.show()
```



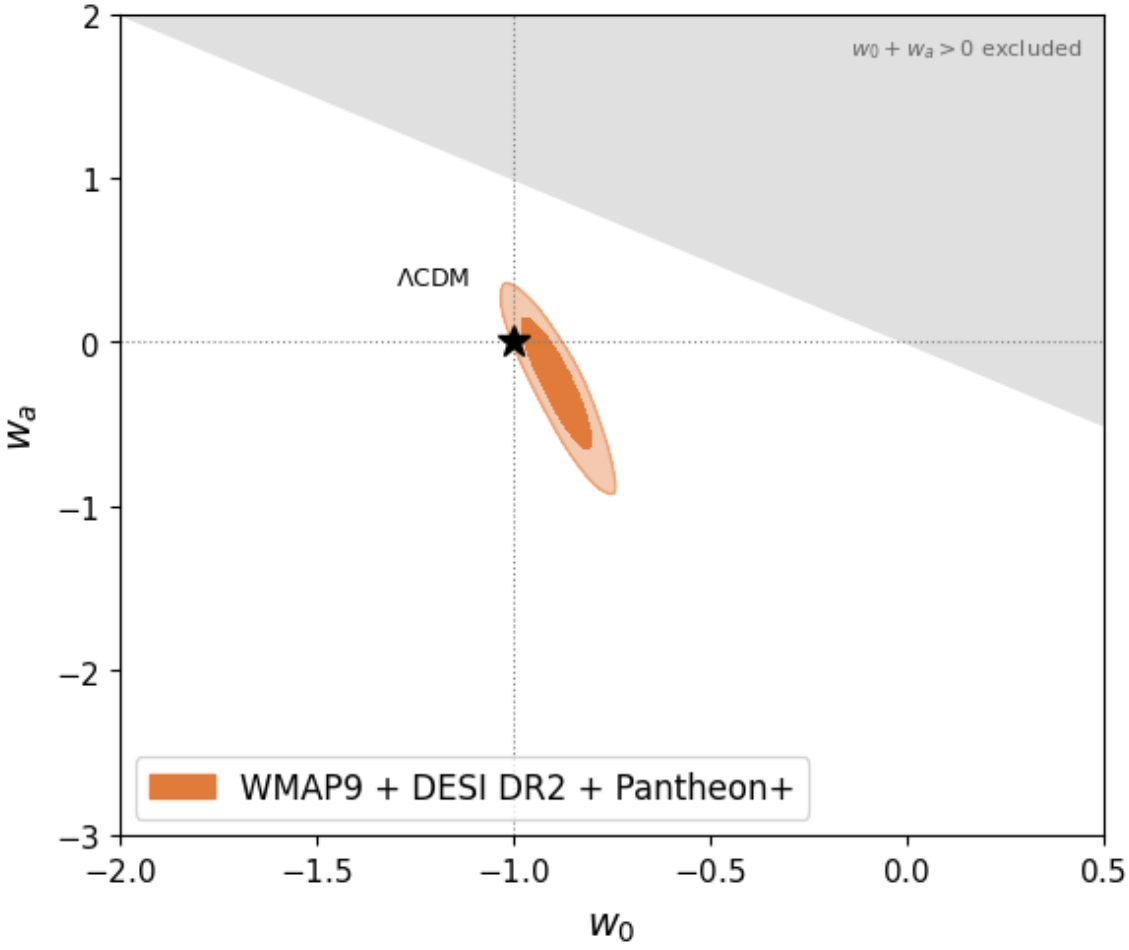
Cell 14 : The w_0–w_a plane

68% and 95% contours with the ΛCDM point marked.
Also, the grey region $w_0 + w_a > 0$ is excluded by the prior.

In [18]:
```
samples, colours = [gd], ['#E07B39']
gd.label = DATA_LABEL
other_tag = 'wmap9_dr2' if USE_SN else 'wmap9_dr2_pantheonplus'
other_root = CHAIN_DIR + f'/{other_tag}_w0wa'
if os.path.exists(other_root + '.1.txt'):
    gd_o = loadMCSamples(other_root, settings={'ignore_rows': 0.3})
    gd_o.label = 'WMAP9 + DESI DR2' + ('' if USE_SN else ' + Panthe
    samples.insert(0, gd_o); colours.insert(0, '#4C72B0')

g = plots.get_single_plotter(width_inch=6, ratio=0.85)
g.plot_2d(samples, 'w', 'wa', filled=True, colors=colours, lims=[-2
ax = g.subplots[0, 0]
x = np.linspace(-2.0, 0.5, 50)
ax.fill_between(x, -x, 2.0, color='0.88', zorder=0)
ax.text(0.45, 1.75, r'$w_0+w_a>0$ excluded', ha='right', fontsize=8
```

```
ax.axvline(-1, color='0.5', lw=0.8, ls=':'); ax.axhline(0, color='0
ax.plot(-1, 0, marker='*', color='k', ms=12, zorder=10)
ax.annotate(r'$\Lambda$CDM', (-1, 0), xytext=(-1.3, 0.35), fontsize
g.add_legend([s.label for s in samples], legend_loc='lower left')
g.export(IMG_DIR + f'/{TAG}_w0_wa.pdf')
g.export(IMG_DIR + f'/{TAG}_w0_wa.png')
plt.show()
```



Cell 15 : Comparison with the WMAP9 posteriors

The existing WMAP9-alone and WMAP9 + DESI DR1 ΛCDM chains copied from `reference_chains/`, in the H_0–Ω_m plane and for the shared ΛCDM parameters.

In [19]:
```
names_ref  = ['ombh2', 'omch2', 'H0', 'ns', 'logA', 'tau', 'omegam'
labels_ref = [LATEX[n] for n in names_ref[:-1]] + [r'\Omega_m']

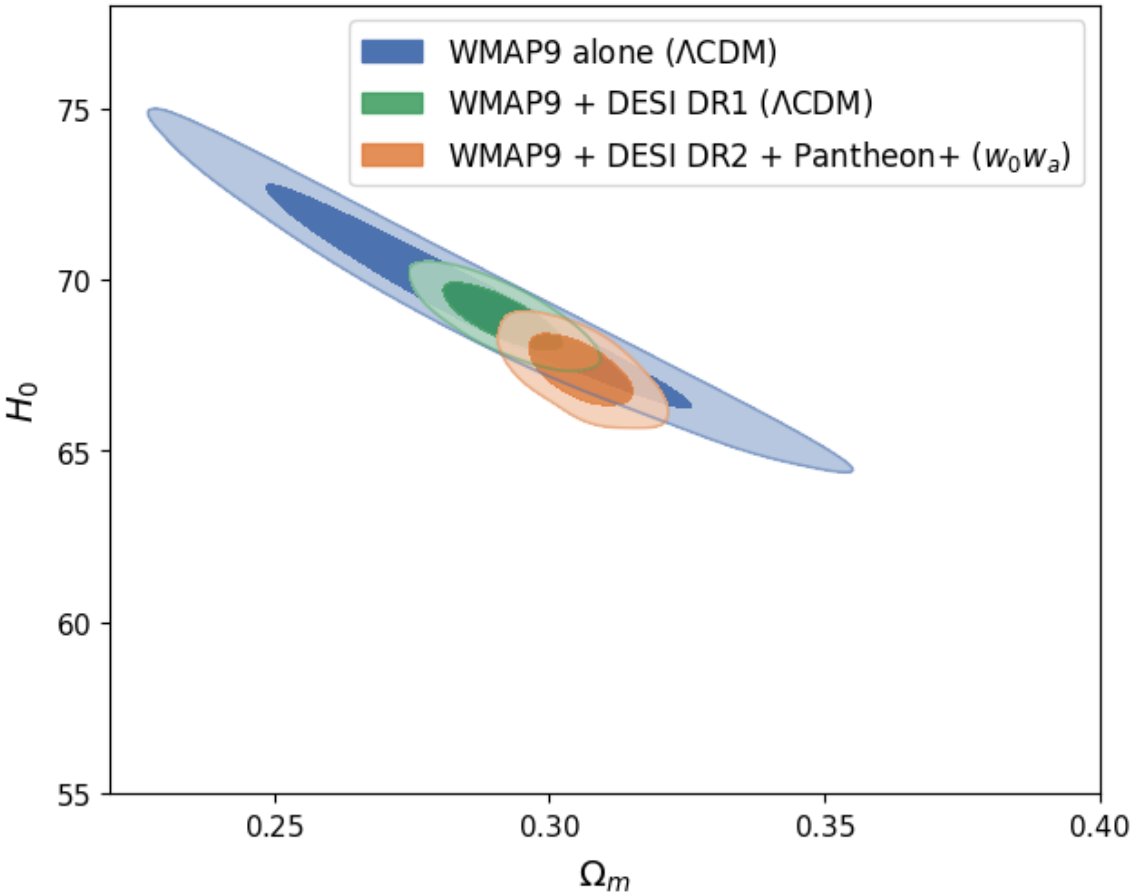
def ref_samples(path, label):
    x  = np.load(path)
    Om = (x[:, 0] + x[:, 1] + 0.06 / 93.14) / (x[:, 2] / 100)**2
    return MCSamples(samples=np.column_stack([x, Om]), names=names_
                     settings={'smooth_scale_1D': 0.5, 'smooth_scal

wmap_alone = ref_samples(REF_DIR + '/wmap9_lcdm_flat_samples_202606
```

```
joint_dr1  = ref_samples(REF_DIR + '/wmap9_desi_joint_corrected_fla
                         r'WMAP9 + DESI DR1 ($\Lambda$CDM)')
gd.label = DATA_LABEL + r' ($w_0w_a$)'

g = plots.get_single_plotter(width_inch=6.5, ratio=0.8)
g.plot_2d([wmap_alone, joint_dr1, gd], 'omegam', 'H0', filled=True,
          colors=['#4C72B0', '#3A9B5C', '#E07B39'], lims=[0.22, 0.4
g.add_legend([wmap_alone.label, joint_dr1.label, gd.label], legend_
g.export(IMG_DIR + f'/{TAG}_H0_Om_vs_wmap.png')
plt.show()

print(f'{"":8s} {"WMAP9 alone":>20s} {"WMAP9+DR1 (LCDM)":>20s} {"th
for p in ['ombh2', 'omch2', 'H0', 'ns', 'logA', 'tau', 'omegam']:
    vals = [s.mean(p) for s in (wmap_alone, joint_dr1, gd)]
    errs = [s.std(p)  for s in (wmap_alone, joint_dr1, gd)]
    print(f'{p:8s} ' + ' '.join(f'{v:11.4f} ± {e:6.4f}' for v, e in
```



```
                  WMAP9 alone     WMAP9+DR1 (LCDM)      this run (w0
wa)
ombh2          0.0226 ± 0.0005      0.0225 ± 0.0004      0.0225 ± 0.0
004
omch2          0.1142 ± 0.0045      0.1151 ± 0.0020      0.1153 ± 0.0
024
H0            69.3946 ± 2.1541     68.9053 ± 0.6513     67.3341 ± 0.7
004
ns             0.9739 ± 0.0130      0.9720 ± 0.0102      0.9716 ± 0.0
103
logA           3.0954 ± 0.0307      3.0962 ± 0.0294      3.0961 ± 0.0
295
tau            0.0887 ± 0.0142      0.0877 ± 0.0131      0.0873 ± 0.0
133
omegam         0.2868 ± 0.0260      0.2914 ± 0.0071      0.3056 ± 0.0
063
```

Cell 16 : Final Summary

In [20]:

```
print('=' * 65)
print(f'  {DATA_LABEL} — Joint w0waCDM Final Summary (Cobaya)')
print('=' * 65)
print(f'  Likelihoods : {", ".join(model.likelihood)}')
print(f'  Samples     : {gd.numrows} (after 30% burn-in)')
print(f'  Final R-1   : {prog[-1]["Rminus1"]:.4f}' if prog.size and
print()
for p in PARAM_NAMES + ['omegam']:
    print(f'  {p:<8} {gd.mean(p):10.5f} ± {gd.std(p):8.5f}')
print(f'  {"w_p":<8} {wp_mean:10.5f} ± {wp_std:8.5f}   (z_p = {1/a_
print()
print(f'  chi2_min LCDM = {float(best["lcdm"]["chi2"]):.2f}   chi2_
print(f'  Delta chi2    = {dchi2:.2f}  ->  {sigma:.2f} sigma prefer
print('=' * 65)
```

```
================================================================
  WMAP9 + DESI DR2 + Pantheon+ — Joint w0waCDM Final Summary (Cobay
a)
================================================================
  Likelihoods : wmap9, bao.desi_dr2.desi_bao_all, sn.pantheonplus
  Samples     : 11648 (after 30% burn-in)
  Final R-1   : 0.0096

  ombh2       0.02255 ±  0.00043
  omch2       0.11534 ±  0.00237
  H0         67.33410 ±  0.70039
  ns          0.97159 ±  0.01032
  logA        3.09615 ±  0.02948
  tau         0.08734 ±  0.01331
  w          -0.88576 ±  0.05852
  wa         -0.25446 ±  0.25972
  omegam      0.30561 ±  0.00627
  w_p        -0.93624 ±  0.02776   (z_p = 0.247)

  chi2_min LCDM = 8975.45   chi2_min w0wa = 8969.52
  Delta chi2    = 5.93  ->  1.95 sigma preference for w0wa (2 extra
parameters)
================================================================
```